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Inventor(s)	Nitta; Jumpei et al.

Image forming apparatus having detachable toner cartridges

Abstract

An image forming apparatus includes a rotatable rotary including a developing roller and a storage frame including a storage portion to store toner, a body frame with an opening, a toner cartridge which is attachable to and detachable from the rotary through the opening, which is movable to an attached position and a retracted position, a moving device to move the toner cartridge from the attached position to the retracted position, and an opening/closing member provided to be movable between a closed position and an open position. The opening/closing member moves from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the moving device.

Inventors: Nitta; Jumpei (Kanagawa, JP), Tateishi; Tomoya (Kanagawa, JP), Sasaoka; Osamu (Kanagawa, JP), Koguchi; Kohei (Kanagawa, JP), Hiasa; Shuma (Kanagawa, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 1000008780869

Assignee: CANON KABUSHIKI KAISHA (Tokyo, JP)

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an image forming apparatus that forms an image on a recording material.

Description of the Related Art

(2) In an image forming apparatus of an electrophotographic system, a rotary developing system that forms a color image by rotating a rotary including a plurality of developing rollers is known. Japanese Patent Application Laid-Open Nos. 2007-183305 and 2008-096852 each disclose an image forming apparatus including a rotary including a plurality of developing rollers, and a plurality of toner cartridges (toner storage containers) each attachable to and detachable from the rotary.

SUMMARY OF THE INVENTION

(3) According to one aspect of the invention, an image forming apparatus includes a rotary configured to rotate and including a developing roller and a storage frame including a storage portion configured to store toner to be supplied to the developing roller, a body frame with an opening, the body frame being configured to accommodate the rotary, a toner cartridge which is configured to store the toner, which is attachable to and detachable from the rotary through the opening, which is movable to an attached position and a retracted position where the toner cartridge is retracted from the attached position, with respect to the storage frame, and which is configured to supply the toner to the storage portion in a state in which the toner cartridge is at the attached position, a moving device configured to move the toner cartridge from the attached position to the retracted position, and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the moving device.

(4) According to another aspect of the invention, an image forming apparatus includes a rotary including a first developing roller, a second developing roller, a first storage frame, and a second storage frame, the first storage frame including a first storage portion configured to store first toner to be supplied to the first developing roller, the second storage frame including a second storage portion configured to store second toner to be supplied to the second developing roller, a body frame configured to accommodate the rotary and having an opening, a first toner cartridge which is configured to store the first toner, which is attachable to and detachable from the rotary through the opening, which is movable to a first attached position and a first retracted position where the first toner cartridge is retracted from the first attached position, with respect to the first storage frame, and which is configured to supply the first toner to the first storage portion in a state in which the first toner cartridge is at the first attached position, a second toner cartridge which is configured to store the second toner, which is attachable to and detachable from the rotary through the opening, which is movable to a second attached position and a second retracted position where the second toner cartridge is retracted from the second attached position, with respect to the second storage

frame, and which is configured to supply the second toner to the second storage portion in a state in which the second toner cartridge is at the second attached position, a first moving device configured to move the first toner cartridge from the first attached position to the first retracted position, and a second moving device configured to move the second toner cartridge from the second attached position to the second retracted position, and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the first toner cartridge and the second toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the first toner cartridge from the first attached position to the first retracted position by the first moving device, and wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the second toner cartridge from the second attached position to the second retracted position by the moving device.

(5) According to another aspect of the invention, an image forming apparatus includes a developing device including a developing roller and a storage frame including a storage portion configured to store toner to be supplied to the developing roller, a body frame with an opening, the body frame being configured to accommodate the developing device, a toner cartridge which is configured to store the toner, which is attachable to and detachable from the developing device through the opening, which is movable between an attached position and a retracted position where the toner cartridge is retracted from the attached position, with respect to the storage frame, and which is configured to supply the toner to the storage portion in a state in which the toner cartridge is at the attached position, a moving device including a support member configured to support the toner cartridge, the moving device being configured to move the toner cartridge from the attached position to the retracted position by movement of the support member, and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the toner cartridge to and from the developing device through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the movement of the support member.

(6) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of an image forming apparatus according to a first embodiment.

(2) FIG. 2 is a configuration diagram of the image forming apparatus according to the first embodiment.

(3) FIG. 3 is a schematic view of a developing unit, a toner cartridge, and a tray according to the first embodiment.

(4) FIGS. 4A and 4B are each a section view of the image forming apparatus according to the first embodiment.

(5) FIG. 5 is a perspective view of a rotary body according to the first embodiment.

(6) FIGS. 6A to 6C are each a perspective view of the image forming apparatus according to the first embodiment.

- (7) FIGS. 7A and 7B are each a section view of the image forming apparatus according to the first embodiment.
- (8) FIG. 8 is an explanatory diagram of the rotary body according to the first embodiment.
- (9) FIG. 9 is an explanatory diagram of the rotary body according to the first embodiment.
- (10) FIG. 10 is an explanatory diagram of the rotary body according to the first embodiment.
- (11) FIGS. 11A and 11B are each an explanatory diagram of elements related to the movement of the tray according to the first embodiment.
- (12) FIGS. 12A and 12B are each an explanatory diagram of elements related to the movement of the tray according to the first embodiment.
- (13) FIG. 13 is a configuration diagram of an image forming apparatus according to a second embodiment.
- (14) FIGS. 14A to 14C are each an explanatory diagram of elements related to the movement of a tray according to the second embodiment.
- (15) FIG. 15 is an explanatory diagram of elements related to the movement of a tray according to a third embodiment.
- (16) FIGS. 16A to 16C are each an explanatory diagram of elements related to the movement of the tray according to the third embodiment.
- (17) FIGS. 17A and 17B are each an explanatory diagram of elements related to the movement of a tray according to a fourth embodiment.
- (18) FIG. 18 is a schematic view of an image forming apparatus according to a fifth embodiment.
- (19) FIGS. 19A and 19B is a schematic view of an image forming apparatus according to a sixth embodiment.
- (20) FIGS. 20A and 20B are each an explanatory diagram illustrating a moving device according to a modification example.

(21) FIG. 21 is a schematic view of an image forming apparatus according to a fifth embodiment.

DESCRIPTION OF THE EMBODIMENTS

(22) Embodiments of the present disclosure will be described below with reference to drawings.

First Embodiment

(23) An image forming apparatus **1** according to a first embodiment will be described with reference to FIGS. 1 to 12B. In the description below and each drawing, the vertical direction in the case where the image forming apparatus **1** is disposed on a horizontal surface will be referred to as a Z direction. A direction that intersects with the Z direction and that is the direction of a rotational axis **90C** of a rotary body **90** (rotational axis direction of a rotary) that will be described later will be referred to as a Y direction. A direction intersecting with both the Z direction and the Y direction will be referred to as an X direction. The X direction and the Y direction are preferably horizontal directions. In addition, the X direction, the Y direction, and the Z direction are preferably orthogonal to each other. In addition, the sides pointed by arrows X, Y, and Z illustrated in each diagram will be respectively referred to as +X, +Y, and +Z sides, and sides opposite thereto will be respectively referred to as -X, -Y, and -Z sides.

(24) Overall Configuration of Image Forming Apparatus

(25) First, an overall configuration of the image forming apparatus **1** will be described. The image forming apparatus **1** is a laser beam printer that forms an image on a sheet **S** by using an electrophotographic method. Specifically, the image forming apparatus **1** is a color laser beam printer including four developing units **50y**, **50m**, **50c**, and **50k**. As the sheet **S** serving as a recording material (recording medium), various sheet materials of different sizes and different materials can be used. Examples of the various sheet materials include paper sheets such as plain paper sheets and cardboards, plastic films, cloths, surface-treated sheet materials such as coated paper sheets, and sheet materials of irregular shapes such as envelopes and index paper sheets.

(26) A schematic configuration of the image forming apparatus **1** and an image forming operation will be described with reference to FIGS. 1, 2, and 3. FIG. 1 is a schematic diagram illustrating a

sectional configuration of the image forming apparatus **1**. FIG. **2** is a diagram for describing drive sources of the image forming apparatus **1**. FIG. **3** is a concept diagram illustrating elements for supplying toner from a toner cartridge **70** to a developing unit **50**.

(27) As illustrated in FIG. **1**, the image forming apparatus **1** includes an image forming apparatus main body (hereinafter referred to as an apparatus body **1A**), and toner cartridges **70y**, **70m**, **70c**, and **70k** that are attachable to and detachable from the apparatus body **1A**. The apparatus body **1A** of the present embodiment is a part of the image forming apparatus **1** excluding the toner cartridges **70y**, **70m**, **70c**, and **70k**.

(28) The apparatus body **1A** of the image forming apparatus **1** includes a photosensitive member **2** for use in an electrophotographic system. The photosensitive member has a drum shape (cylindrical shape) and is hereinafter referred to as a photosensitive drum **2**. The photosensitive member **2** serves as an image bearing member that bears an electrostatic latent image. A charging roller **3**, a scanner **4** serving as an exposing device, and a cleaning unit **6** are disposed around the photosensitive drum **2**.

(29) The charging roller **3** is an example of a charging means or a charging unit for uniformly charging the photosensitive drum **2**. The scanner **4** is an example of an exposing means or an exposing unit that exposes the photosensitive drum **2** by irradiating the photosensitive drum **2** with laser light in accordance with image information. By irradiating the photosensitive drum **2** with the laser light after charging, an electrostatic latent image is formed on each surface of the photosensitive drum **2**. The cleaning unit **6** is an example of a cleaning means or a cleaning portion that removes toner remaining on the surface of the photosensitive drum **2**.

(30) Further, the apparatus body **1A** includes a sheet storage portion **300**, a pickup roller **310**, a feed roller **311**, a separation roller **312**, a conveyance roller pair **320**, a secondary transfer roller **12**, a fixing device **40**, and an intermediate transfer unit **10**. The pickup roller **310** is an example of a feeding means or a feeding unit that feeds the sheet **S**. The feed roller **311** and the separation roller **312** are an example of a separation conveyance unit that conveys sheets **S** while separating the sheets **S** from each other by frictional force. The secondary transfer roller **12** is an example of a transfer means or a transfer unit that transfers an image from an intermediate transfer belt **10a** onto the sheet **S**.

(31) The intermediate transfer unit **10** includes an intermediate transfer belt **10a**, a belt driving roller **10b**, a tension roller **10c**, a cleaning device **13**, and a primary transfer roller **11**. The intermediate transfer belt **10a** is an example of an intermediate transfer member that bears an image transferred from the photosensitive drum **2** through primary transfer and conveys the image so as to transfer the image onto the sheet **S** through secondary transfer. The intermediate transfer belt **10a** is stretched over the belt driving roller **10b** and the tension roller **10c**. The belt driving roller **10b** is a driving member that is rotationally driven by a drive source to convey the intermediate transfer belt **10a**.

(32) In addition, the apparatus body **1A** includes a rotary body **90**, which serves as a rotary, rotator, or developing device, including the developing units **50y**, **50m**, **50c**, and **50k**. As will be described later, trays **80y**, **80m**, **80c**, and **80k**, which serve as support members, are attached to the rotary body **90** in the present embodiment. Toner cartridges **70y**, **70m**, **70c**, and **70k** are detachably attached to the trays **80y**, **80m**, **80c**, and **80k**.

(33) In the description below, a plurality of members having similar functions can be distinguished by numbers given thereto. For example, one of the toner cartridges **70y**, **70m**, **70c**, and **70k** may be referred to as a first toner cartridge, one of the remaining three may be referred to as a second toner cartridge, one of the remaining two may be referred to as a third toner cartridge, and the last one may be referred to as a fourth toner cartridge. Similarly, one of the trays **80y**, **80m**, **80c**, and **80k** may be referred to as a first tray, one of the remaining three may be referred to as a second tray, one of the remaining two may be referred to as a third tray, and the last one may be referred to as a fourth tray. That is, one of the trays **80y** to **80k** is an example of a first support member, another one

of the trays **80y** to **80k** is an example of a second support member, yet another one of the trays **80y** to **80k** is an example of a third support member, and the last one of the trays **80y** to **80k** is an example of a fourth support member. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

(34) The developing units **50y**, **50m**, **50c**, and **50k** serving as first to fourth developing units are examples of developing means or developing portions that each develop (visualize) an electrostatic latent image formed on the photosensitive drum **2** into a toner image by using toner of a corresponding color. The developing units **50y**, **50m**, **50c**, and **50k** each develop the electrostatic latent image formed on the photosensitive drum **2** by using corresponding one of yellow toner, magenta toner, cyan toner, and black toner. The developing units **50y**, **50m**, **50c**, and **50k** may be arranged in an order different from the order illustrated in FIG. **1**.

(35) The developing unit **50y** includes a developing roller **51y**, a supply roller **52y**, and a developing blade. The developing roller **51y** is a developer bearing member that rotates while bearing toner serving as developer or developing agent, and supplies the toner to the photosensitive drum **2**. The supply roller **52y** is a supply member that is disposed in contact with the developing roller **51y** and supplies toner to the developing roller **51y**. The developing blade is a regulation member that regulates the thickness of a toner layer borne on the developing roller **51y**. The other developing units **50m**, **50c**, and **50k** respectively include developing rollers **51m**, **51c**, and **51k**, supply rollers, and developing blades that are configured in a similar manner.

(36) The toner cartridges **70y**, **70m**, **70c**, and **70k** corresponding to the developing units **50y**, **50m**, **50c**, and **50k** are attached to the rotary body **90**. The toner cartridges **70y**, **70m**, **70c**, and **70k** respectively store therein yellow toner, magenta toner, cyan toner, and black toner to be supplied to the developing units **50y**, **50m**, **50c**, and **50k**. One of the toners of four colors may be referred to as first toner, one of the toners of remaining three colors may be referred to as second toner, one of the toners of remaining two colors may be referred to as third toner, and the toner of the last remaining color may be referred to as fourth toner. For example, the black toner may be referred to as an example of first toner, and the magenta toner may be referred to as an example of second toner. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

(37) Here, the rotary body **90** includes a rotary frame **90f** supporting the developing units **50y**, **50m**, **50c**, and **50k**. The developing units **50y**, **50m**, **50c**, and **50k** are supported by the rotary frame **90f** that is a rotary support member that is rotatable.

(38) In addition, the trays **80y**, **80m**, **80c**, and **80k** are attached to the rotary body **90**. The rotary body **90** and the trays **80y**, **80m**, **80c**, and **80k** as a combination can be referred to as a rotary unit **90U**. In other words, the rotary unit **90U** includes the rotary body **90** and the trays **80y**, **80m**, **80c**, and **80k**.

(39) The toner cartridges **70y** to **70k** are detachably held by the trays **80y** to **80k**. As will be described later, the trays **80y** to **80k** are supported so as to be slidable to the outside of the rotary body **90**. The rotary unit **90U** and the toner cartridges **70y**, **70m**, **70c**, and **70k** as combined can be referred to as a rotary assembly **90A**. In other words, the rotary assembly **90A** includes the rotary unit **90U** and toner cartridges **70y**, **70m**, **70c**, and **70k**.

(40) As will be described later, the rotary body **90** is rotatable about a rotational axis (rotational center) **90C**. The rotational axis **90C** coincides with a rotational axis of the rotary frame **90f**, that of the rotary unit **90U**, and that of the rotary assembly **90A**. In addition, the rotational axis **90C** is substantially parallel to the rotational axis (rotational center) of the photosensitive drum **2**.

(41) The rotary body **90** rotates about the rotational axis **90C**, and thus can take developing postures in each of which any one of the developing rollers **51y**, **51m**, **51c**, and **51k** faces the photosensitive drum **2**. A posture in which the developing roller **51y** faces the photosensitive drum **2** will be referred to as a yellow developing posture. A posture in which the developing roller **51m** faces the photosensitive drum **2** will be referred to as a magenta developing posture. A posture in

which the developing roller **51c** faces the photosensitive drum **2** will be referred to as a cyan developing posture. A posture in which the developing roller **51k** faces the photosensitive drum **2** will be referred to as a black developing posture. That is, the rotary body **90** can rotate about the rotational axis **90C** such that the positions of the developing rollers **51y**, **51m**, **51c**, and **51k** change with respect to the photosensitive drum **2**. The black developing posture is an example of a first developing posture in which the first developing roller (e.g., developing roller **51k**) faces the photosensitive drum **2**. The other developing postures are examples of a second developing posture in which a second developing roller (e.g., one of the developing rollers **51y** to **51c**) faces the photosensitive drum **2**. The yellow/magenta/cyan/black developing postures can be referred to as first to fourth developing postures. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

(42) As illustrated in FIG. 2, the apparatus body **1A** includes motors **M1**, **M2**, and **M3** serving as drive sources. As will be described later, the motor **M1** supplies a driving force for rotating the rotary body **90** about the rotational axis **90C**. In other words, the motor **M1** rotates the rotary assembly **90A** and the rotary unit **90U** about the rotational axis **90C**.

(43) In addition, the apparatus body **1A** includes a driving device **98** including the motor **M2** and a transmission device. The transmission device includes driving racks **15L** and **15R** serving as driving gears and a transmission portion **15t** that will be described later. The driving force of the motor **M2** is transmitted to the driving racks **15L** and **15R** by the transmission portion **15t**. In other words, the motor **M2** is configured to drive the driving racks **15L** and **15R**, and moves the trays **80y**, **80m**, **80c**, and **80k** with respect to the rotary body **90** via the driving racks **15L** and **15R**.

(44) The motor **M3** drives members that are not driven by the motors **M1** and **M2**. For example, the motor **M3** drives the photosensitive drum **2**, the developing units **50y**, **50m**, **50c**, and **50k**, the pickup roller **310**, the feed roller **311**, the conveyance roller pair **320**, the secondary transfer roller **12**, the belt driving roller **10b**, and the fixing device **40**.

(45) To be noted, the members driven by the motors **M1**, **M2**, and **M3** can be appropriately changed. In addition, the roles of two or three of the motors **M1**, **M2**, and **M3** can be concentrated in one motor. In addition, a drive source other than the motors **M1**, **M2**, and **M3** may be added.

(46) Here, the suffices **y**, **m**, **c**, and **k** given to the developing units **50y**, **50m**, **50c**, and **50k**, the toner cartridges **70y**, **70m**, **70c**, and **70k**, the trays **80y**, **80m**, **80c**, and **80k**, and the like indicate the colors of toner. The developing units **50y**, **50m**, **50c**, and **50k** basically have the same configuration and function. The toner cartridges **70y**, **70m**, **70c**, and **70k** basically have the same configuration and function. In addition, the trays **80y**, **80m**, **80c**, and **80k** basically have the same configuration and function. Therefore, in the case where these do not need to be distinguished, the suffices **y**, **m**, **c**, and **k** will be omitted, and arbitrarily selected one of the four units, four cartridges, and four trays will be described.

(47) As illustrated in FIG. 3, the toner cartridge **70** includes a toner frame **71**. The toner frame **71** includes a toner storage portion **71a** that stores toner, and a discharge opening **71b** communicating with the toner storage portion **71a**.

(48) The developing unit **50** includes a developing frame **53** serving as a storage frame. The developing frame **53** includes a developing-side storage portion **53a** and an inlet opening **53b** communicating with the developing-side storage portion (toner supply chamber) **53a**. To be noted, as described above, although the developing unit **50** includes the developing roller **51**, the supply roller **52**, and the like, illustration of these members is omitted in FIG. 3.

(49) The developing roller **51k** included in the developing unit **50k** is an example of a first developing roller. The developing roller **51m** included in the developing unit **50m** is an example of a second developing roller. A developing frame **53k** of the developing unit **50k** including the developing-side storage portion **53a** illustrated in FIG. 4A is an example of a first storage frame including a first storage portion. A developing frame **53m** of the developing unit **50m** including the developing-side storage portion **53a** illustrated in FIG. 4A is an example of a second storage frame

including a second storage portion. The rotary body **90** is an example of a rotary that is rotatable and includes a first developing roller, a second developing roller, a first storage frame including a first storage portion, and a second storage frame including a second storage portion. In the present embodiment, the rotary body **90** includes first to fourth developing rollers and first to fourth storage frames.

(50) As will be described later, the toner cartridge **70** is movable to an attached position and a retracted position where the toner cartridge **70** is retracted from the attached position, with respect to the developing frame **53**. In a state in which the toner cartridge **70** is at the attached position with respect to the developing frame **53**, the discharge opening **71b** faces the inlet opening **53b**. That is, the toner storage portion **71a** of the toner cartridge **70** and the developing-side storage portion **53a** of the developing unit **50** communicate with each other via the discharge opening **71b** and the inlet opening **53b**. When toner is supplied from the toner cartridge **70** to the developing unit **50**, at least part of the inlet opening **53b** is positioned below at least part of the discharge opening **71b**.

(51) Then, toner stored in the toner storage portion **71a** is discharged through the discharge opening **71b**, and the toner discharged through the discharge opening **71b** is stored in the developing-side storage portion **53a** through the inlet opening **53b**. The toner stored in the developing-side storage portion **53a** is supplied to the developing roller **51** by the supply roller **52**. The toner stored in the toner storage portion **71a** is supplied to the developing roller **51** through such a path.

(52) The toner cartridge **70** preferably includes an unillustrated sealing member (first sealing member) that covers the discharge opening **71b**. In addition, the developing unit **50** preferably includes an unillustrated sealing member (second sealing member) that covers the inlet opening **53b**.

(53) In a state in which the toner cartridge **70** is not attached to the developing unit **50**, the discharge opening **71b** and the inlet opening **53b** are preferably each covered by a sealing member such that leakage of toner through the discharge opening **71b** and the inlet opening **53b** is suppressed.

(54) Image Forming Operation

(55) An image forming operation in the present embodiment will be described. First, the photosensitive drum **2** is rotated in an arrow direction (counterclockwise direction) in FIG. **1** in synchronization with the rotation of the intermediate transfer belt **10a**. Further, the surface of the photosensitive drum **2** is uniformly charged by the charging roller **3**.

(56) In the case of forming a color image on the sheet **S**, the rotary body **90** rotates in an arrow direction (clockwise direction) in FIG. **1** while supporting the developing units **50y**, **50m**, **50c**, and **50k**. Then, an electrophotographic process is repeatedly performed while moving the developing rollers **51y**, **51m**, **51c**, and **51k** to a developing position one by one.

(57) First, the scanner **4** emits laser light based on image data corresponding to a yellow image, and thus forms an electrostatic latent image corresponding to the yellow image on the surface of the photosensitive drum **2**. In parallel with the formation of this electrostatic latent image, the motor **M1** rotates the rotary body **90**, and the rotary body **90** takes the yellow developing posture. When the rotary body **90** is in the yellow developing posture, the developing roller **51y** is in the developing position, and the electrostatic latent image formed on the photosensitive drum **2** is developed with yellow toner.

(58) Here, in the present embodiment, the developing rollers **51y**, **51m**, **51c**, and **51k** are each an elastic roller formed by covering a metal shaft with rubber. At the developing position, the developing rollers **51y**, **51m**, **51c**, and **51k** each develop the electrostatic latent image in a state of being in contact with the photosensitive drum **2**. That is, a contact development system is employed for the image forming apparatus **1** of the present embodiment. However, at the developing position, each of the developing rollers **51y**, **51m**, **51c**, and **51k** may develop the electrostatic latent image with a gap between the developing roller and the photosensitive drum **2**. That is, a non-contact development system may be employed for the image forming apparatus **1**.

(59) After the yellow toner image is developed, the yellow toner image on the photosensitive drum **2** is transferred onto the intermediate transfer belt **10a** through primary transfer by the primary transfer roller **11** disposed on the inner peripheral side of the intermediate transfer belt **10a**.

(60) After this, toner images of respective colors are formed by rotating the rotary body **90** and thus sequentially moving the developing rollers **51m**, **51c**, and **51k** to the developing position. That is, after the yellow toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** takes a magenta developing posture, and a magenta toner image is formed on the intermediate transfer belt **10a**. After the magenta toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** takes a cyan developing posture, and a cyan toner image is formed on the intermediate transfer belt **10a**. After the cyan toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** takes a black developing posture, and a black toner image is formed on the intermediate transfer belt **10a**. After the black toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** rotates about the rotational axis **90C** in an arrow direction (clockwise direction) illustrated in FIG. **1**, and returns to the yellow developing posture. To be noted, the color of the image to be formed first on the intermediate transfer belt **10a** can be arbitrarily selected, and for example, the black toner image may be formed first.

(61) Then, primary transfer is repeated so as to superimpose the toner images of four colors on each other on the intermediate transfer belt **10a**, and thus a color image is formed on the intermediate transfer belt **10a**. To be noted, before the color image is formed on the intermediate transfer belt **10a**, the secondary transfer roller **12** and the cleaning device **13** are not in contact with the intermediate transfer belt **10a**.

(62) Meanwhile, the sheet **S** is fed by the pickup roller **310** from the sheet storage portion **300** provided in a lower portion of the apparatus body **1A**. The sheet **S** is conveyed to the conveyance roller pair **320** in a state in which one sheet **S** is separated from a stack of sheets **S** by the feed roller **311** and the separation roller **312**. The conveyance roller pair **320** delivers out the fed sheet **S** to a transfer portion (secondary transfer portion) that is a nip portion between the intermediate transfer belt **10a** and the secondary transfer roller **12**. The color image on the intermediate transfer belt **10a** is transferred onto the surface of the conveyed sheet **S** through secondary transfer.

(63) The sheet **S** onto which a color image has been transferred is conveyed to the fixing device **40**. In the fixing device **40**, the sheet **S** is heated and pressurized, and thus the image is fixed to the sheet **S**. The sheet **S** having passed the fixing device **40** is discharged to the outside of the image forming apparatus **1** as a product.

(64) In contrast, in the case of forming a black-and-white image (monochrome image) on the sheet **S**, the rotary body **90** takes the black developing posture. In this state, an electrostatic latent image is formed on the surface of the photosensitive drum **2** by charging and exposing the photosensitive drum **2**, and then the electrostatic latent image is developed with black toner by the developing roller **51k** positioned at the developing position. The black toner image is transferred onto the intermediate transfer belt **10a** through primary transfer, and then the toner image is transferred onto the sheet **S** through secondary transfer. Steps after this are similar to the case of a color image.

(65) Rotary Configuration

(66) The configuration of the rotary body **90** will be described with reference to FIGS. **1**, **4A**, **4B**, and **5**. FIGS. **4A** and **4B** are each a section view of the rotary body **90** of the image forming apparatus **1** and the surroundings thereof. To be noted, FIGS. **4A** and **4B** are each a section view taken along a virtual plane orthogonal to the rotational axis **90C** of the rotary body **90**. FIG. **5** is a perspective view of the rotary body **90**.

(67) As has been described, the toner cartridges **70y** to **70k** are attachable to and detachable from the rotary body **90**. In the case where toner in the toner cartridges **70y** to **70k** has run out, the user can replenish the image forming apparatus **1** with toner by replacing the toner cartridges **70y** to **70k**.

(68) As illustrated in FIG. **1**, the apparatus body **1A** includes a frame **16** accommodating the rotary

body **90**. The frame **16** is a body frame of the image forming apparatus **1** of the present embodiment. The frame **16** is a casing or skeleton of the apparatus body **1A** constituted by a frame and exterior members, and has an approximately rectangular parallelepiped shape.

(69) The frame **16** has an opening **16a**. More specifically, the frame **16** has a side surface **16b** extending in a direction intersecting with the horizontal direction. The side surface **16b** constitutes at least part of the exterior surface of the apparatus body **1A** on the +X side. The opening **16a** is provided in the side surface **16b**. The side surface **16b** is a side surface disposed on the downstream side of the discharge port in a discharge direction in which the sheet S on which an image has been formed is discharged from the apparatus body **1A** through the discharge port. From the side surface **16b** side of the image forming apparatus **1**, the user can access the sheet storage portion **300** to replenish the sheet storage portion **300** with sheets S, and can obtain the sheet S discharged through the discharge port. Therefore, the side surface **16b** can be referred to as the front surface of the apparatus body **1A**.

(70) The toner cartridges **70y**, **70m**, **70c**, and **70k** are attachable to and detachable from the rotary body **90** through the opening **16a**. That is, the toner cartridge **70k** can be referred to as an example of a first toner cartridge that stores toner to be supplied to the first developing roller (developing roller **51k**) and that is attachable to and detachable from the rotary (rotary body **90**) through the opening **16a** of the frame **16** of the apparatus body **1A**. The toner cartridge **70m** can be referred to as an example of a second toner cartridge that stores toner to be supplied to the second developing roller (developing roller **51m**) and that is attachable to and detachable from the rotary (rotary body **90**) through the opening **16a** of the frame **16** of the apparatus body **1A**.

(71) In the present embodiment, the toner cartridges **70y**, **70m**, **70c**, and **70k** are attached to and detached from the rotary body **90** through the opening **16a** in the state of being supported by the trays **80y** to **80k**. In other words, the user can attach and detach the toner cartridges **70y** to **70k** to and from the rotary body **90** via the trays **80y** to **80k**.

(72) The opening **16a** is disposed in the side surface **16b** of the frame **16**. In the present embodiment, the side surface **16b** is a surface approximately parallel to the rotational axis **90C** of the rotary body **90**. Therefore, in the case of replacing the toner cartridge **70**, the toner cartridge **70** passes through the opening **16a** in a direction intersecting with the rotational axis **90C** (preferably a direction orthogonal to the rotational axis **90C**).

(73) The image forming apparatus **1** includes a door **14** that covers the opening **16a** of the frame **16**. The door **14** is an opening/closing member movable to a closed position illustrated in FIG. 6A where the door **14** covers the opening **16a** and an open position illustrated in FIGS. 6B and 6C where the opening **16a** is exposed.

(74) As described above, in the present embodiment, the toner cartridge **70** is configured to be attachable to and detachable from the rotary body **90** via the tray **80**. Therefore, the toner cartridge **70** can be stably attached to and detached from the rotary body **90**.

(75) More specifically, the user can replace the toner cartridge **70** by an operation of attaching and detaching the toner cartridge **70** to and from the tray **80** configured to be movable with respect to the rotary body **90** (that is, with respect to the apparatus body **1A**). In the case of a configuration in which the toner cartridge is replaced by the user directly inserting and removing the toner cartridge in and from the apparatus body, the user is required to insert the toner cartridge to a predetermined attached position in the apparatus body. In the present embodiment, the tray **80** is capable of moving such that the toner cartridge **70** moves to the attached position in a state in which the tray **80** is supporting the toner cartridge **70**. Therefore, the user can replace the toner cartridge **70** by a simple operation of placing the toner cartridge **70** on the tray **80**, and thus the operability is improved.

(76) To be noted, the toner cartridge **70** has a thin elongated shape with the Y direction parallel to the rotational axis **90C** of the rotary body **90** as a longitudinal direction. That is, the dimension of the toner cartridge **70** in the longitudinal direction is larger than its height and width in a cross-

section orthogonal to the longitudinal direction. In the case of using the toner cartridge **70** having a thin elongated shape as described above, the toner cartridge **70** can be passed through the opening **16a** in a short movement distance by providing the opening **16a** in the side surface **16b** of the frame **16** that is approximately parallel to the longitudinal direction (Y direction) of the toner cartridge **70**. The replacement of the toner cartridge **70** becomes easier than, for example, a case where the toner cartridge **70** is inserted or removed through an opening provided in a side surface on one side (+Y side or -Y side) of the frame **16** in the longitudinal direction of the toner cartridge **70**.

(77) In the present embodiment, the four toner cartridges **70y**, **70m**, **70c**, and **70k** can be replaced via the opening **16a** that is a shared opening by rotating the rotary body **90**. Therefore, there is no need to provide a plurality of openings respectively for the toner cartridges in the frame **16**, and the total area of the opening provided in the frame **16** for toner cartridge replacement can be reduced. Since the total area of the opening in the frame **16** is smaller, the strength of the frame **16** can be improved. In addition, since only one opening **16a** is provided as a user interface for attaching and detaching a plurality of toner cartridges **70**, the user is not confused about where to attach or detach the toner cartridge **70**, and thus the replacement work can be easily performed.

(78) The rotary body **90** rotates about the rotational axis **90C**, and thus can take a replacement posture in which detachment of one of the toner cartridges **70y** to **70k** from the rotary body **90** is allowed. A posture in which the detachment of the toner cartridge **70y** is allowed will be referred to as a yellow replacement posture. A posture in which the detachment of the toner cartridge **70m** is allowed will be referred to as a magenta replacement posture. A posture in which the detachment of the toner cartridge **70c** is allowed will be referred to as a cyan replacement posture. A posture in which the detachment of the toner cartridge **70k** is allowed will be referred to as a black replacement posture. The black replacement posture is an example of a first replacement posture in which detachment of a first toner cartridge from the rotary body **90** is allowed. The yellow/magenta/cyan replacement postures are examples of a second replacement posture in which detachment of a second toner cartridge from the rotary body **90** is allowed. The yellow/magenta/cyan/black replacement postures can be referred to as first to fourth replacement postures. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

(79) The rotary body **90** rotates about the rotational axis **90C** in a counterclockwise direction of FIG. **1**, and can sequentially take the yellow/magenta/cyan/black replacement postures. In the present embodiment, the rotary body **90** rotates about the rotational axis **90C** in the counterclockwise direction of FIG. **1**, and thus the developing posture and the replacement posture can be switched alternately. For example, in FIG. **1**, the rotary body **90** is in the black developing posture. By rotating the rotary body **90** in the clockwise direction from this state, the posture of the rotary body **90** can be switched in the order of the cyan replacement posture, the yellow developing posture, the black replacement posture, the magenta developing posture, the yellow replacement posture, the cyan developing posture, and the magenta replacement posture. By rotating the rotary body **90** in the clockwise direction from the magenta replacement posture, the rotary body **90** returns to the black developing posture. That is, the rotary body **90** can rotate more than once (360°) in the clockwise direction.

(80) FIG. **4A** illustrates a cross-section of the rotary body **90** in a developing posture (specifically, the yellow developing posture). FIG. **4B** illustrates a cross-section of the rotary body **90** in a replacement posture (specifically, the black replacement posture).

(81) As illustrated in FIGS. **4A** and **4B**, the four trays **80y** to **80k** are attached to the rotary body **90**. The trays **80y** to **80k** respectively hold the toner cartridges **70y** to **70k**. In FIGS. **4A** and **4B**, the trays **80y** to **80k** are accommodated in the rotary body **90**, and this state can be referred to as a state in which the toner cartridges **70y** to **70k** are attached to the developing units **50y**, **50m**, **50c**, and **50k**.

(82) As described above, the toner cartridge **70** is movable to an attached position and a retracted position where the toner cartridge **70** is retracted from the attached position, with respect to the developing frame **53** of the developing unit **50**. That is, the first toner cartridge (toner cartridge **70k**) is movable to a first attached position and a first retracted position with respect to a first storage frame (developing frame **53k**). The second toner cartridge (toner cartridge **70m**) is movable to a second attached position and a second retracted position with respect to a second storage frame (developing frame **53m**).

(83) In a state in which the toner cartridge **70** is at the attached position with respect to the developing frame **53**, the discharge opening **71b** and the inlet opening **53b** face each other as illustrated in FIG. 3. The toner cartridge **70** is configured to supply toner to the developing-side storage portion **53a** through the inlet opening **53b** (opening of the storage frame) in this state.

(84) The apparatus body **1A** includes a moving device **85** configured to move the toner cartridge **70** from the attached position to the retracted position with respect to the rotary body **90** (more specifically, with respect to the developing frame **53** of the developing unit **50**). The moving device **85** will be described below with reference to FIG. 8 and the like. In the present embodiment, a plurality of moving devices **85y** to **85k** corresponding to the plurality of toner cartridges **70y** to **70k** are disposed in the rotary body **90**. The trays **80y** to **80k** can be referred to as part of the moving devices **85y** to **85k**.

(85) In the present embodiment, the toner cartridge **70k** storing the black toner is larger in size than the toner cartridges **70y** to **70c** storing yellow toner, magenta toner, and cyan toner, and is capable of storing more toner. In other words, the first toner cartridge is capable of storing a first amount of toner, the second toner cartridge is capable of storing a second amount of toner, and the first amount is larger than the second amount.

(86) Specifically, the length of the black toner cartridge **70k** in a first radial direction with respect to the rotational axis **90C** of the rotary body **90** is larger than the length of the magenta toner cartridge **70m** in a second radial direction. Here, the first radial direction is a rotational radius direction (radial direction of a virtual circle centered on the rotational axis **90C**) of the rotary body **90**, and is a direction in which the toner cartridge **70k** extends with respect to the rotational axis **90C** as viewed in the direction of the rotational axis **90C**. The second radial direction is a rotational radius direction of the rotary body **90**, and is a direction in which the toner cartridge **70m** extends with respect to the rotational axis **90C** as viewed in the direction of the rotational axis **90C**. Similarly, the length of the black toner cartridge **70k** in the first radial direction is larger than the lengths of the other toner cartridges **70y** and **70c** in the radial directions corresponding to the toner cartridges **70y** and **70c**.

(87) Accordingly, the tray **80k** holding the black toner cartridge **70k** is larger in size than the trays **80y** to **80c** holding the other toner cartridges **70y**, **70m**, and **70c**. That is, the four toner cartridges **70y** to **70k** and the trays **80y** to **80k** having different sizes are disposed in the rotary body **90**. In other words, the toner cartridge **70k** serving as an example of a first toner cartridge and the toner cartridge **70y** serving as an example of a second toner cartridge smaller than the first toner cartridge are attachable to and detachable from the rotary body **90**. In accordance with this, the tray **80k** serving as an example of a first support member that supports the first toner cartridge and the tray **80y** serving as an example of a second support member smaller than the first support member are provided in the rotary body **90**. In addition, the toner cartridges **70m** and **70c** serving as examples of a third toner cartridge and a fourth toner cartridge that are smaller in size than the first toner cartridge are attachable to and detachable from the rotary body **90**. In accordance with this, the trays **80m** and **80c** serving as examples of a third support member and a fourth support member that are smaller in size than the first support member are provided in the rotary body **90**.

(88) Here, rotational driving of the rotary body **90** will be described with reference to FIG. 5. As illustrated in FIG. 5, disk gears **92L** and **92R** are formed on respective end portions of the rotary body **90**. In addition, rotary driving gears **93L** and **93R** are formed on respective end portions of a

swing shaft **91** in a manner capable of transmitting a driving force. Here, the driving force of the motor **M1** is transmitted to the rotary driving gear **93R** via a drive transmission mechanism. Next, the driving force is transmitted to the disk gears **92L** and **92R** via the rotary driving gears **93L** and **93R**, and thus the rotary body **90** is rotationally driven. The rotary body **90** rotates about the rotational axis **90C** in the clockwise direction in FIG. 1.

(89) In addition, the rotary body **90** is supported so as to be swingable about the swing shaft **91**. The rotary body **90** is urged in a counterclockwise direction in FIGS. 4A and 4B about the swing shaft **91** by an unillustrated urging member. This direction can be referred to as a direction in which the developing rollers **51y** to **51k** each approach the photosensitive drum **2**. As a result of this, one of the developing rollers **51y** to **51k** is in contact with the photosensitive drum **2** in a state in which the rotary body **90** is in a developing posture.

(90) Meanwhile, as illustrated in FIG. 5, rotary cams **90eL** and **90eR** are provided on respective end portions of the rotary body **90**. When the rotary body **90** rotates about the rotational axis **90C** in a clockwise direction in FIGS. 4A and 4B, the rotary cams **90eL** and **90eR** come into contact with a roller **96** supported by the frame **16** illustrated in FIGS. 4A and 4B. Then, the rotary body **90** moves in the clockwise direction in FIGS. 4A and 4B about the swing shaft **91**. This direction can be referred to as a direction in which the developing rollers **51y** to **51k** each move away from the photosensitive drum **2**. In addition, this direction can be referred to as a direction in which the rotary body **90** approaches the opening **16a** of the frame **16** and the door **14**.

(91) As a result of this, when the rotary body **90** rotates and switches from the developing posture to the replacement posture, the rotary body **90** swings about the swing shaft **91**. In a state in which the rotary body **90** is in the replacement posture, the developing roller **51** is separated from the photosensitive drum **2**.

(92) As illustrated in FIG. 4B, in the black replacement posture, the toner cartridge **70k** stops at a position where the toner cartridge **70k** faces the opening **16a** and the door **14** provided on the side surface **16b** of the apparatus body **1A**. When the tray **80k** is slid from the attached position for the developing unit **50k** to the outside of the rotary body **90** from this state, the user can replace the toner cartridge **70k**.

(93) Replacement Operation of Toner Cartridge

(94) A toner cartridge replacement operation will be described with reference to FIGS. 4A, 6A to 6C, 7A, and 7B. FIGS. 6A to 6C are each an exterior view of the apparatus body **1A**. FIGS. 7A and 7B are each a section view of the rotary body **90** and the surroundings thereof in toner cartridge replacement. To be noted, FIGS. 7A and 7B are each a section view of the apparatus taken along a virtual plane orthogonal to the rotational axis **90C** of the rotary body **90**.

(95) FIG. 6A illustrates an external appearance of the apparatus body **1A** during the image forming operation and in a standby state. During the image forming operation is during a period in which a series of operations of the image forming apparatus **1** feeding a sheet **S**, forming an image on the sheet **S**, and then discharging the sheet **S** as a product are executed. The standby state is a state in which the image forming operation can be started if the image forming apparatus **1** receives an image forming instruction (printing instruction), and a state in which the image forming apparatus **1** is standing by for the image forming instruction from the user. As illustrated in FIG. 6A, the door **14** is closed during the image forming operation and in the standby state.

(96) FIG. 6B illustrates the external appearance of the apparatus body **1A** at the time of toner cartridge replacement. At the time of toner cartridge replacement, the door **14** is opened, and the tray **80** and the toner cartridge **70** are moved to the outside of the apparatus body **1A**.

(97) The toner cartridge **70** is movable to an attached position and a retracted position where the toner cartridge **70** is retracted from the attached position, with respect to the developing frame **53** of the developing unit **50**. In a state in which the toner cartridge **70** is at the attached position with respect to the developing frame **53**, the discharge opening **71b** and the inlet opening **53b** face each other as illustrated in FIG. 3. As illustrated in FIGS. 4A and 4B, the rotary body **90** is configured to

rotate about the rotational axis **90C** to take the developing posture and the replacement posture in a state in which the toner cartridge **70** is at the attached position.

(98) The toner cartridge replacement operation will be described. First, the user instructs the controller of the apparatus body **1A** of a toner cartridge replacement operation. The instruction of toner cartridge replacement operation is given by, for example, input via an operation panel (operation portion) provided on the apparatus body **1A**.

(99) When the controller receives the instruction of the toner cartridge replacement operation, the rotary body **90** rotates to the replacement posture of the toner cartridge **70** serving as a replacement target (toner cartridge **70** whose toner has run out), and stops. That is, the controller rotates the rotary body **90** to the replacement posture of a toner cartridge specified in the instruction of the toner cartridge replacement (in FIG. **4B**, the black replacement posture for replacing the black toner cartridge **70k**). In the replacement posture, the tray **80** supporting the toner cartridge **70** whose replacement has been instructed faces the opening **16a** of the frame **16** of the apparatus body **1A**.

(100) For example, the rotary body **90** of FIG. **4A** is in the yellow developing posture in which the yellow developing roller **51y** faces the photosensitive drum **2**. At this time, the black toner cartridge **70k** and the tray **80k** do not have to face the opening **16a** and the door **14**. In other words, the toner cartridge **70** and the tray **80** do not have to face the opening **16a** and the door **14** in the case where the rotary body **90** is in a developing posture or a replacement posture other than the replacement posture of the toner cartridge. Therefore, the opening **16a** may have such a size that each of the toner cartridges **70** can individually pass therethrough. When the rotary body **90** rotates in the clockwise direction in the drawings by a predetermined angle from the yellow developing posture, the black toner cartridge **70k** and the tray **80k** face the opening **16a** and the door **14** as illustrated in FIG. **4B**.

(101) Here, “the tray **80** facing the opening **16a**” means that the tray **80** is positioned so as to be able to move to the outside of the apparatus body **1A** through the opening **16a**. That is, in the case where the tray **80** faces the opening **16a**, a moving mechanism that will be described later moves the tray **80** outward in the rotational radius direction of the rotary body **90**, and therefore the tray **80** and the toner cartridge **70** supported by the tray **80** are capable of projecting to the outside of the apparatus body **1A**. In FIG. **4A**, none of the trays **80y** to **80k** faces the opening **16a**. In FIG. **4B**, only the black tray **80k** faces the opening **16a**, and the other trays **80y** to **80c** do not face the opening **16a**.

(102) When the rotary body **90** is positioned in the replacement posture, the motor **M2** moves the tray **80** supporting the toner cartridge **70** serving as a replacement target to the outside of the apparatus body **1A**. As a result of this, the toner cartridge **70** serving as a replacement target moves from the attached position to the retracted position with respect to the rotary body **90**. In addition, as illustrated in FIGS. **6B**, **6C**, **7A**, and **7B**, the tray **80** and the toner cartridge **70** serving as a replacement target supported by the tray **80** projects to the outside of the apparatus body **1A** through the opening **16a**.

(103) More specifically, the tray **80** is movable to an accommodation position and a detachment position with respect to the rotary body **90**. The accommodation position is a position where the tray **80** is accommodated in the rotary body **90**. The detachment position is a position (removal position or replaceable position) where the tray **80** projects to the outside of the rotary body **90** and the toner cartridge **70** can be detached from the tray **80**. The positions of the trays **80y** to **80k** in FIGS. **4A** and **4B** serve as examples of the accommodation position. The position of the tray **80** in FIGS. **6B** and **6C**, the position of the tray **80k** in FIG. **7A**, and the position of the tray **80m** in FIG. **7B** serve as examples of the detachment position.

(104) When the tray **80** is at the accommodation position, the toner cartridge **70** attached to the tray **80** is at the attached position. When the tray **80** is at the detachment position, the toner cartridge **70** attached to the tray **80** is at the retracted position.

(105) Here, as illustrated in FIGS. **7A** and **7B**, the rotary body **90** has projection portions **95** for

holding the tray **80** at the accommodation position and holding the toner cartridge **70** at the attached position. As illustrated in FIG. **8**, the tray **80** has recess portions **87** configured to be fit on the projection portions **95**. FIGS. **7A** and **7B** illustrate projection portions **95k** and **95m** corresponding to the trays **80k** and **80m**, and FIG. **8** illustrates recess portions **87y** and **87m** of the trays **80y** and **80m**. The projection portion **95** and the recess portion **87** are provided for each of the trays **80y** to **80k**. The projection portion **95** is preferably urged in such a direction as to engage with the recess portion **87**.

(106) The projection portion **95** fits in the recess portion **87** of the tray **80**, and thus the tray **80** is locked with respect to the rotary frame **90f**. As a result of this, the tray **80** stays in the accommodation position even when the rotary body **90** rotates, and thus movement of the toner cartridge **70** from the attached position can be suppressed. To be noted, in the case where the tray **80** is moved between the accommodation position and the detachment position by the moving device that will be described below, the projection portion **95** can be configured to be moved by the tray **80** and thus disengaged from the recess portion **87**.

(107) In the present embodiment, the door **14** is supported so as to be pivotable with respect to the apparatus body **1A**. As illustrated in FIG. **7A**, the door **14** is urged from the open position to the closed position by a spring **14s**. The spring **14s** is, for example, a tension spring, and urges the door **14** such that a moment in a counterclockwise direction in FIGS. **7A** and **7B** is generated about a support shaft **14c** of the door **14**.

(108) The tray **80** pushes the door **14**, and thus the door **14** takes the open state illustrated in FIG. **6B**. This state can be referred to as a state in which the tray **80** is supported by the door **14**. The door **14** supports at least part of the tray **80** projecting to the outside of the apparatus body **1A**, and thus the toner cartridge **70** can be supported more stably. In other words, when the first toner cartridge (toner cartridge **70k**) is at the first retracted position, the opening/closing member (door **14**) at the open position supports the first support member (tray **80k**). In addition, when the second toner cartridge (one of the toner cartridges **70y** to **70c**) is at the second retracted position, the opening/closing member (door **14**) at the open position supports the second support member (one of the trays **80y** to **80c**).

(109) To be noted, the door **14** is configured such that the door **14** at the open position comes in contact with part (for example, a lower edge **16c** of the opening **16a**) of the frame **16** of the apparatus body **1A** and not to pivot downward beyond the open position. When the tray **80** is pulled back into the apparatus body **1A** from the outside, the door **14** returns to the closed position by the urging force of the spring **14s**.

(110) The toner cartridge **70** is detachably held by the tray **80**. Therefore, as illustrated in FIG. **6C**, the user can perform a work (replacement work) of detaching the toner cartridge **70** from the tray **80** and attaching a new toner cartridge **70**. To be noted, in the case of replacing a plurality of toner cartridges **70**, the replacement work can be performed by repeating the operation described above.

(111) As described above, in the present embodiment, a configuration in which the opening/closing member (e.g., door **14**) moves from the closed position to the open position in an interconnected manner with movement of the toner cartridge **70** from the attached position to the retracted position by the moving device **85** has been employed. As a result of this, in the present embodiment, the moving device **85** includes a support member (e.g., tray **80**) supporting the toner cartridge **70**, and the opening/closing member (e.g., door **14**) moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **70** from the attached position to the retracted position by the movement of the support member. As a result of this, the user does not have to open the door **14** manually, which makes it easier for the user to replace the toner cartridge.

(112) In addition, in the present embodiment, since the opening/closing member (e.g., door **14**) is urged toward the closed position by the spring **14s** serving as an urging member, the opening/closing member moves from the open position to the closed position in an interconnected

manner with the movement of the toner cartridge **70** from the retracted position to the attached position by the moving device **85**. As a result of this, the user does not have to close the door **14** manually after replacing the toner cartridge, which makes it easier for the user to replace the toner cartridge.

(113) In addition, in the present embodiment, in accordance with the movement of the toner cartridge **70** from the attached position to the retracted position, the support member (e.g., tray **80**) pushes the opening/closing member (e.g., door **14**), and thus the door **14** moves from the closed position to the open position. As a result of this, movement of the opening/closing member interconnected with the movement of the toner cartridge **70** can be realized without providing an additional member for moving the opening/closing member. To be noted, the opening/closing member can be also moved in an interconnected manner with the movement of the toner cartridge **70** by a configuration different from the configuration in which the support member pushes the opening/closing member as in a modification example that will be described later.

(114) FIGS. **7A** and **7B** illustrate a cross-section of the rotary body **90** and the surroundings thereof at the time of toner cartridge replacement. FIG. **7A** illustrates a state at the time of replacing the black toner cartridge **70k**. FIG. **7B** illustrates a state at the time of replacing the magenta toner cartridge **70m**.

(115) The image forming apparatus **1** includes the moving device **85** illustrated in FIG. **8** that moves the toner cartridge **70** from the attached position to the retracted position. In the present embodiment, it can be said that the moving device **85** includes the tray **80**. A moving device **85k** including the tray **80k** can be referred to as an example of a first moving device including a first support member. A moving device **85m** including the tray **80m** can be referred to as an example of a second moving device including a second support member.

(116) Even when the toner cartridge **70** is at the retracted position, the tray **80** is coupled to the rotary body **90** (supported by the rotary body **90**). To easily detach the toner cartridge **70** from the rotary body **90**, it is preferable that the length by which the toner cartridge **70** projects from the rotary body **90** at the retracted position is large. Since the toner cartridge **70** is configured to be attachable to and detachable from the rotary body **90** via the tray **80**, the toner cartridge **70** can be stably supported by the tray **80** even in the case where the length by which the toner cartridge **70** projects from the rotary body **90** is large.

(117) The movement direction of the toner cartridge **70** in which the toner cartridge **70** moves from the attached position to the retracted position will be referred to as a retraction direction. In the present embodiment, the retraction direction of the toner cartridge **70** is a direction intersecting with the direction of the rotational axis **90C** (Y direction). Therefore, as illustrated in FIGS. **7A** and **7B**, as viewed in the direction of the rotational axis **90C** (Y direction), the retraction direction of the toner cartridge **70** is a direction orthogonal to the direction of the rotational axis **90C** (Y direction). In addition, the retraction direction of the toner cartridge **70** can be referred to as a direction outward in the rotational radius direction of the rotary body **90**, that is, a direction away from the rotational axis **90C**.

(118) As illustrated in FIGS. **7A** and **7B**, since the user detaches the toner cartridge **70** from the rotary body **90**, it is preferable that the at least part of the toner cartridge **70** projects from the rotary body **90** at the time of detaching the toner cartridge **70**. In the present embodiment, the entirety of the toner cartridge **70** projects from the rotary body **90** when the toner cartridge **70** is at the retracted position.

(119) It can be said that when the rotary body **90** rotates about the rotational axis **90C**, the rotational trajectory of the rotary body **90** matches a circumscribed circle of the rotary body **90** centered on the rotational axis **90C**, that is, a virtual circle **90V** indicated by a broken line in FIGS. **7A** and **7B**. When the toner cartridge **70** is at the retracted position, half the length or more of the toner cartridge **70** in the retracted direction is preferably outside the rotational trajectory of the rotary body **90**. That is, as viewed in the rotational axis direction of the rotary, half the total length

or more of the toner cartridge is preferably positioned outside the rotational trajectory of the rotary in the movement direction of the toner cartridge from the attached position to the retracted position. This applies to the toner cartridges **70** including the toner cartridge **70k** serving as an example of the first cartridge and the toner cartridge **70m** serving as an example of the second cartridge. In addition, in the present embodiment, the entirety of the toner cartridge **70** is outside the rotational trajectory (virtual circle **90V**) of the rotary body **90** when the toner cartridge **70** is at the retracted position as illustrated in FIGS. **7A** and **7B**.

(120) Further, to make it easier for the user to grab the toner cartridge **70**, at least part of the toner cartridge **70** is preferably outside the image forming apparatus **1**, that is, outside the apparatus body **1A** when the toner cartridge **70** is at the retracted position. The outside of the apparatus mentioned herein refers to a space that is outside the image forming apparatus **1** (outside the apparatus body **1A**) when the image forming apparatus **1** is used for, for example, an image forming operation on the sheet **S**.

(121) In the present embodiment, the exterior surface of the apparatus body **1A** is constituted by the exterior surface of the frame **16**. That is, outside the apparatus can be also referred to as outside the frame **16**. Therefore, a state in which at least part of the toner cartridge **70** is outside the apparatus can be also referred to as a state in which at least part of the toner cartridge **70** projects to the outside of the frame **16** through the opening **16a** of the frame **16** of the apparatus body **1A**.

(122) In the present embodiment, when the door **14** is at the closed position, the opening **16a** of the frame **16** of the apparatus body **1A** is covered by the door **14**. Further, the exterior surface **14a** of the door **14** at the closed position constitutes part of the exterior surface of the apparatus body **1A**. In this case, outside of the apparatus refers to the outside of the exterior surface **14a** of the door **14** at the closed position. That is, in the case where the position of the exterior surface **14a** of the door **14** at the closed position is referred to as an exterior position, at least part of the toner cartridge **70** is positioned more outward than the exterior position with respect to the apparatus body **1A** when the toner cartridge **70** is at the retracted position.

(123) In other words, at least part of the toner cartridge **70** is positioned in a space that would be outside the apparatus body **1A** if the door **14** were at the closed position. Further, at least part of the toner cartridge **70** is positioned downstream of the exterior position in the retraction direction of the toner cartridge **70**.

(124) In addition, in the case where the side surface **16b** having the opening **16a** is the front surface of the apparatus body **1A**, it can be said at least part of the toner cartridge **70** projects to the front side more than the exterior surface on the front side of the apparatus body **1A** when the toner cartridge **70** is at the retracted position. In this case, the user can easily access the toner cartridge **70** from the front side of the image forming apparatus and replace the toner cartridge **70**.

(125) To be noted, when the toner cartridge **70** is at the retracted position, half the length or more of the toner cartridge **70** in the retraction direction is preferably outside the apparatus. That is, as viewed in the rotational axis direction of the rotary, half the total length or more of the toner cartridge is preferably positioned outside the body frame in the movement direction of the toner cartridge from the attached position to the retracted position in a state in which the toner cartridge is at the retracted position. This applies to the toner cartridges **70** including the toner cartridge **70k** serving as an example of the first toner cartridge and the toner cartridge **70m** serving as an example of the second toner cartridge. In addition, the entirety of the toner cartridge **70** is preferably outside the apparatus when the toner cartridge **70** is at the retracted position. To be noted, although the exterior surface **14a** of the door **14** and the side surface **16b** constitute the exterior surface on the front side of the apparatus body **1A** in the present embodiment, the configuration of the door **14** is not limited to this. For example, the size of the door **14** may be set so as to cover the entirety of the side surface **16b**. In this case, the exterior surface **14a** of the door **14** constitutes the exterior surface on the front side of the apparatus body **1A**.

(126) The tray **80** includes a cartridge holding portion **81** illustrated in FIGS. **3** and **6C** that holds

the toner cartridge **70**. The cartridge holding portion **81** is an attached portion to which the toner cartridge **70** is attached. When the tray **80** is at the detachment position, the entirety of the cartridge holding portion **81** is preferably outside the rotational trajectory of the rotary body **90** in the retraction direction. When the tray **80** is at the detachment position, half the length or more of the cartridge holding portion **81** is preferably outside the apparatus in the retraction direction.

(127) Here, as described above, the toner cartridge **70k** and the tray **80k** are larger in size than the other toner cartridges **70y** to **70c** and the other trays **80y** to **80c**. Therefore, as illustrated in FIGS. 7A and 7B, in the present embodiment, the movement amount of the tray **80** in toner cartridge replacement is changed in accordance with the size of the toner cartridge **70**.

(128) Specifically, as illustrated in FIG. 7A, the movement distance of the tray **80k** (first support member) from the accommodation position (first accommodation position) to the detachment position (first detachment position) is **L1**. The movement distance of the tray **80m** (second support member) from the accommodation position to the detachment position (third detachment position) is **L2**. Although a state in which the toner cartridge **70m** and the tray **80m** are moved is illustrated in FIG. 7B, the movement distance of the trays **80y** and **80c** from the accommodation position to the detachment position is also **L2**. In this case, **L1** is larger than **L2**. In other words, it can be said that the movement distance of the first support member in the case where the first toner cartridge moves from the first attached position to the first retracted position is larger than the movement distance of the second support member in the case where the second toner cartridge moves from the second attached position to the second retracted position.

(129) In addition, as illustrated in FIG. 7A, in a state in which the tray **80k** is at the detachment position and the toner cartridge **70k** is at the retracted position, the toner cartridge **70k** projects from the exterior surface of the apparatus body **1A** to the outside of the apparatus by a distance **P1**. In the present embodiment, the tray **80k** also projects from the exterior surface of the apparatus body **1A** to the outside of the apparatus by the distance **P1**.

(130) In addition, as illustrated in FIG. 7B, in a state in which the tray **80m** is at the detachment position and the toner cartridge **70m** is at the retracted position, the toner cartridge **70m** projects from the exterior surface of the apparatus body **1A** to the outside of the apparatus by a distance **P2**. In the present embodiment, the tray **80m** also projects from the exterior surface of the apparatus body **1A** to the outside of the apparatus by the distance **P2**. To be noted, the toner cartridges **70y** and **70c** also project from the exterior surface of the apparatus body **1A** to the outside of the apparatus by the distance **P2**.

(131) The distance **P1** described above is larger than the distance **P2**. That is, the length by which the first toner cartridge at the first retracted position projects through the opening **16a** of the apparatus body **1A** will be referred to as a first length (**P1**), and the length by which the second toner cartridge at the second retracted position projects through the opening **16a** will be referred to as a second length (**P2**). In this case, it can be said that the first length is larger than the second length.

(132) It is more preferable in terms of strength that the distance **P2** by which the toner cartridges **70y** to **70c** smaller in size than the toner cartridge **70k** project to the outside of the apparatus at the retracted position is smaller than the distance **P1** by which the toner cartridge **70k** projects to the outside at the retracted position. The reason for this is as follows. When the toner cartridge **70** is at the retracted position, at least part of the toner cartridge **70** projects to the outside of the apparatus from the outside of the rotational trajectory of the rotary body **90** or the exterior surface of the apparatus body **1A**. At this time, the tray **80** supports the weight of the toner cartridge **70** in a state in which one side thereof is supported by the rotary body **90**. Therefore, reducing the distance **P2** by which the toner cartridges **70y** to **70c** project to the outside of the apparatus at the retracted position can reduce the load on the trays **80y** to **80c** or guide portions **97** of the rotary body **90** supporting the trays **80y** to **80k**. In addition, since the toner cartridges **70y** to **70c** are smaller than the toner cartridge **70k**, the operability in the cartridge replacement for the trays **80y** to **80c** can be

maintained even if the distance P2 is set to be smaller than the distance P1.

(133) Tray Arrangement in Rotary

(134) The arrangement of the trays **80y** to **80k** in the rotary body **90** will be described with reference to FIGS. **8**, **9**, and **10**. FIG. **8** is a perspective view illustrating the arrangement of the trays **80y** to **80k** in the rotary body **90**. FIG. **9** is a section view illustrating the arrangement of the trays **80y** to **80k** in the rotary body **90**. FIG. **10** is a diagram illustrating the member arrangement of the trays **80y** to **80k** on one end side in the Y direction. To be noted, FIG. **9** illustrates a cross-section of the rotary body **90** taken along a virtual plane orthogonal to the rotational axis **90C** of the rotary body **90**. In addition, the top half of FIG. **10** is a diagram of the rotary body **90** and the trays **80m** and **80k** of FIG. **8** as viewed from the upper-right side (+Z side) of FIG. **8**, and the lower half of FIG. **10** is a diagram of the rotary body **90** and the trays **80c** and **80y** of FIG. **8** as viewed from the left side (-X) side of FIG. **8**.

(135) As illustrated in FIG. **8**, the trays **80y** to **80k** are respectively provided with cartridge holding portions **81y** to **81k** and guided portions **82y** to **82k**.

(136) The toner cartridges **70y** to **70k** are respectively attached to the cartridge holding portions **81y** to **81k**. The cartridge holding portions **81y** to **81k** respectively accommodate at least part of the toner cartridges **70y** to **70k** attached thereto.

(137) The guided portions **82y** to **82k** are provided at respective end portions of the trays **80y** to **80k** such that the cartridge holding portions **81y** to **81k** are each interposed therebetween in the Y direction. The guided portions **82y** to **82k** are each a thin and long member extending in a direction orthogonal to the rotational axis of the rotary body **90**.

(138) In the present embodiment, a reinforcing rib **82k1** is formed on part of the guided portion **82k** in a movement direction Dk of the tray **80k**, and a reinforcing rib **82m1** is formed on part of the guided portion **82m** in a movement direction Dm of the tray **80m** as illustrated in FIGS. **11A** and **11B**. The reinforcing ribs **82k1** and **82m1** are thin and long rib shapes (ridges) projecting outward in the Y direction from the guided portions **82k** and **82m** of the respective end portions of the trays **80k** and **80m** in the Y direction and extending in the movement directions Dk and Dm of the trays **80k** and **80m**. The reinforcing ribs **82k1** and **82m1** improve the stiffness of the guided portions **82k** and **82m**.

(139) To be noted, although the lengths of the reinforcing ribs **82m1** and **82k1** in the present embodiment are restricted so as to avoid the guided portions **82y** and **82c**, the reinforcing ribs **82m1** and **82k1** may be provided for the entire lengths of the guided portions **82m** and **82k** unless interference with the guided portions **82y** and **82c** occurs. Reinforcing ribs may be added to the guided portions **82y** and **82c**. In addition, in the case where the stiffness of the guided portions **82m** and **82k** is sufficient, a configuration in which the reinforcing ribs **82m1** and **82k1** are not provided may be employed.

(140) The guided portions **82y** to **82k** have rack portions (rack gears) **83y** to **83k** formed therein. In addition, pinion gears **94y** to **94k** are rotatably held in the rotary body **90**. The pinion gears **94y** to **94k** are respectively engaged with the rack portions **83y** to **83k** in a manner capable of transmitting a driving force.

(141) The rack portions **83y** to **83k** and the pinion gears **94y** to **94k** are part of the moving devices **85y** to **85k** configured to move the toner cartridges **70y** to **70k** from the attached position to the retracted position. In addition, it can be said that the rack portions **83y** to **83k** and the pinion gears **94y** to **94k** are part of a driven device driven by the driving device **98** of the apparatus body **1A**. It can be said that the pinion gears **94y** to **94k** are rotatable bodies (rotary members) that rotate to move the trays **80y** to **80k** with respect to the rotary body **90**.

(142) The pinion gears **94y** to **94k** and the rack portions **83y** to **83k** function as driven portions for the moving devices **85y** to **85k** of the rotary body **90** to receive the driving force from the driving device **98** of the apparatus body **1A**. The pinion gear **94k** and the rack portion **83k** are examples of a first pinion gear and a first rack gear constituting at least part of a first driven portion included in

a first moving device. The pinion gear **94m** and the rack portion **83m** are examples of a second pinion gear and a second rack gear constituting at least part of a second driven portion included in a second moving device.

(143) The rotary body **90** includes guide portions **97** illustrated in FIGS. 7A and 7B that respectively engage with the guided portions **82y** to **82k**. FIG. 7A illustrates the guide portion **97** (**97k**) that engages with the guided portion **82k** of the tray **80k**, and FIG. 7B illustrates the guide portion **97** (**97m**) that engages with the guided portion **82m** of the tray **80m**. The rotary body **90** includes similar guide portions that respectively engage with the guided portions **82y** and **82c** of the trays **80y** and **80c**. In addition, although the guide portions **97** provided on one side (+Y side) of the rotary body **90** in the Y direction are illustrated in FIGS. 7A and 7B, similar guide portions **97** are also provided on the other side (-Y side) of the rotary body **90** in the Y direction.

(144) When the tray **80** moves between the accommodation position and the detachment position, the guide portion **97** maintains the state of engaging with the guided portion **82** in at least part of the movement range, and guides the movement direction of the tray **80**. In the present embodiment, the guide portion **97** maintains the state of engaging with the guided portion **82k** in the entirety of the movement range of the tray **80k** between the accommodation position and the detachment position. In addition, in the present embodiment, the guide portion **97** maintains the state of engaging with the guided portion **82m** in the entirety of the movement range of the tray **80m** between the accommodation position and the detachment position.

(145) In addition, as illustrated in FIGS. 8 and 9, the four trays **80y** to **80k** are disposed in the rotary body **90** so as to overlap each other as will be described in detail below.

(146) When the pinion gears **94y** to **94k** rotate, the rack portions **83y** to **83k** and the trays **80y** to **80k** move with respect to the rotary body **90**. As illustrated in FIG. 9, the four trays **80y** to **80k** are arranged such that the movement directions thereof with respect to the rotary body **90** are rotated by 90° from each other. Therefore, the tray **80y** and the tray **80c** are held so as to be slidable in substantially the same direction (parallel directions) as each other, and the tray **80m** and the tray **80k** are held so as to be slidable in substantially the same direction (parallel directions) as each other. The movement directions of sliding of the trays **80y** to **80k** are regulated by the engagement between the guide portions **97** and the guided portions **82y** to **82k** described above.

(147) To be noted, the trays **80y** to **80k** move to the outside of the apparatus through the opening **16a**. When the trays **80y** to **80k** each move to the outside of the apparatus through the opening **16a**, the movement directions of the trays are substantially the same (parallel).

(148) As illustrated in FIG. 9, the trays are disposed such that with respect to the movement direction Dk of the tray **80k**, the range in which the tray **80k** is disposed overlaps with the range in which the tray **80y** is disposed and the range in which the tray **80c**. In addition, with respect to the movement direction Dk of the tray **80k**, the range in which the tray **80k** is disposed overlaps with the rotational axis **90C** of the rotary body **90**. That is, it can be said that the toner cartridge **70k** held by the cartridge holding portion **81k** of the tray **80k** overlaps with the rotational axis **90C** of the rotary body **90** illustrated in FIG. 4B.

(149) In contrast, the trays are disposed in a displaced manner such that with respect to the movement direction Dm of the tray **80m**, the range in which the tray **80m** is disposed does not overlap with the range in which the tray **80y** is disposed and the range in which the tray **80c** is disposed. Further, the trays are disposed in a displaced manner such that with respect to a movement direction Dy of the tray **80y**, the range in which the tray **80y** is disposed does not overlap with the range in which the tray **80m** is disposed and the range in which the tray **80k** is disposed. Similarly, the trays are disposed in a displaced manner such that with respect to a movement direction Dc of the tray **80c**, the range in which the tray **80c** is disposed does not overlap with the range in which the tray **80m** is disposed and the range in which the tray **80k** is disposed.

(150) The positional relationship between the trays **80** can be also expressed as follows. As viewed in the movement direction Dy of the tray **80y**, the tray **80y** and the tray **80k** overlap with each other,

but the tray **80y** and the tray **80m** do not overlap with each other. As viewed in the movement direction Dm of the tray **80m**, the tray **80m** and the tray **80k** overlap with each other, but the tray **80m** and the trays **80y** and **80c** do not overlap with each other. As viewed in the movement direction Dc of the tray **80c**, the tray **80c** and the tray **80k** overlap with each other, but the tray **80c** and the tray **80m** do not overlap with each other.

(151) Here, two elements (such as members, parts, and units) overlapping as viewed in a specific direction refers to a projected region of one element and a projected region of the other element at least partially overlapping with each other in the case where the elements are orthogonally projected onto a virtual plane orthogonal to the specific direction.

(152) As illustrated in FIGS. **8** and **10**, with respect to the direction of the rotational axis **90C** (Y direction), a range in which the rack portion **83m** and the guided portion **82m** are disposed and a range in which the rack portion **83k** and the guided portion **82k** are disposed overlap with each other at least partially. That is, in the present embodiment, it can be said that with respect to the rotational axis direction of the rotary (Y direction), a range in which the first rack gear (rack portion **83k**) is disposed and a range in which the second rack gear (rack portion **83m**) is disposed overlap with each other at least partially. Therefore, as compared with a layout in which the rack portion **83m** and the guided portion **82m** do not overlap with the rack portion **83k** and the guided portion **82k**, the rack portions **83m** and **83k** and the guided portions **82m** and **82k** can be arranged in a smaller space in the Y direction.

(153) With respect to the direction of the rotational axis **90C** (Y direction), a range in which the rack portion **83y** and the guided portion **82y** are disposed and a range in which the rack portion **83c** and the guided portion **82c** are disposed overlap with each other at least partially. That is, in the present embodiment, it can be said that with respect to the rotational axis direction of the rotary (Y direction), a range in which the third rack gear (rack portion **83y**) is disposed and a range in which the fourth rack gear (rack portion **83c**) is disposed overlap with each other at least partially. Therefore, as compared with a layout in which the rack portion **83y** and the guided portion **82y** do not overlap with the rack portion **83c** and the guided portion **82c**, the rack portions **83y** and **83c** and the guided portions **82y** and **82c** can be arranged in a smaller space in the Y direction.

(154) Here, an engagement position between the rack portion **83** and the pinion gear **94** will be described with reference to FIG. **10**. A top half portion of FIG. **10** illustrates an engagement position between the rack portion **83k** and the pinion gear **94k**. A lower half portion of FIG. **10** illustrates an engagement position between the rack portion **83y** and the pinion gear **94y**.

(155) In a region Y1 in the drawings in the direction of the rotational axis **90C** of the rotary body **90** (Y direction), the driving force transmitted from the motor M2 illustrated in FIG. **2** serving as a drive source to a transmission device that will be described below is transmitted to the pinion gears **94y** to **94k**. In a region Y2 in the drawings in the Y direction, the pinion gear **94k** is engaged with the rack portion **83k** in a manner capable of transmitting the driving force. In a region Y3 in the drawings in the Y direction, the pinion gear **94y** is engaged with the rack portion **83y** in a manner capable of transmitting the driving force. To be noted, the rack portion **83m** is engaged with the pinion gear **94m** illustrated in FIG. **8** in the region Y2 similarly to the rack portion **83k** in a manner capable of transmitting the driving force. The rack portion **83c** is engaged with the pinion gear **94c** illustrated in FIG. **8** in the region Y3 similarly to the rack portion **83y** in a manner capable of transmitting the driving force.

(156) Here, the regions Y2 and Y3 are at different positions in the Y direction, that is, are displaced from each other in the Y direction. In addition, the region Y1 is at a position different from both the regions Y2 and Y3 in the Y direction. That is, the region Y1 is displaced from the regions Y2 and Y3 in the Y direction.

(157) Further, in a state in which the toner cartridges **70y** and **70c** are at the attached position, with respect to a movement direction of the rack portion **83y** (movement direction Dy of the tray **80y**), a range in which the rack portion **83y** is disposed and a range in which the rack portion **83c** is

disposed overlap with each other at least partially. In the present embodiment, since the movement directions Dy and Dc of the trays **80y** and **80c** are substantially the same (parallel), also with respect to the movement direction Dc of the tray **80c**, a range in which the rack portion **83y** is disposed and a range in which the rack portion **83c** is disposed overlap with each other at least partially. Therefore, in a state in which the toner cartridges **70y** and **70c** are at the attached position, the tooth surface of the rack portion **83y** faces the tooth surface of the rack portion **83c** in a direction (left-right direction of FIG. 8) orthogonal to the movement directions Dy and Dc of the rack portions **83y** and **83c**.

(158) Further, in a state in which the toner cartridges **70m** and **70k** are at the attached position, with respect to a movement direction of the rack portion **83m** (movement direction Dm of the tray **80m**), a range in which the rack portion **83m** is disposed and a range in which the rack portion **83k** is disposed overlap with each other at least partially. In the present embodiment, since the movement directions Dm and Dk of the trays **80m** and **80k** are substantially the same (parallel), also with respect to the movement direction Dk of the tray **80k**, a range in which the rack portion **83m** is disposed and a range in which the rack portion **83k** is disposed overlap with each other at least partially. Therefore, in a state in which the toner cartridges **70m** and **70k** are at the attached position, the tooth surface of the rack portion **83m** faces the tooth surface of the rack portion **83k** in a direction (up-down direction of FIG. 8) orthogonal to the movement directions Dm and Dk of the rack portions **83m** and **83k**.

(159) In addition, as also illustrated in FIG. 12A that will be described later, the rack portion **83y** overlaps with the rack portions **83m** and **83k** as viewed in the direction (Y direction) of the rotational axis **90C**. As viewed in the direction (Y direction) of the rotational axis **90C**, the rack portion **83m** overlaps with the rack portions **83y** and **83c**. As viewed in the direction (Y direction) of the rotational axis **90C**, the rack portions **83c** and **83m** overlap with the rack portion **83k**. As viewed in the direction of the rotational axis **90C** (Y direction), the rack portions **83k** and **83y** overlap with the rack portion **83c**. In other words, it can be said that with respect to the rotational axis direction of the rotary (Y direction), a range in which the first rack gear (rack portion **83k**) is disposed does not overlap with a range in which the second rack gear (rack portion **83y**) is disposed. In addition, it can be said that, as viewed in the rotational axis direction of the rotary (Y direction), in a state in which the first toner cartridge **70k** is at the first attached position and the second toner cartridge **70y** is at the second attached position, the first rack gear (rack portion **83k**) and the second rack gear (rack portion **83y**) overlap with each other.

(160) As described above, since the positions where the rack portions **83k** and **83m** are disposed and the positions where the rack portions **83y** and **83c** are disposed are different in the Y direction, the rack portions can be arranged such that the rack portions **83y** and **83c** overlap with the rack portions **83m** and **83k** as viewed in the Y direction.

(161) As a result of this, the space for arranging the four trays in the rotary body **90** can be reduced, and thus the size of the rotary body **90** in the rotational radius direction can be reduced. That is, when it is attempted to arrange the rack portions **83** such that the movement distances of the trays **80y** to **80k** are equivalent to those of the present embodiment and the rack portions **83** do not overlap with each other as viewed in the Y direction, the area required for the arrangement of the four rack portions is large as viewed in the Y direction. As compared with such a configuration, as a result of arranging the plurality of rack portions **83** in a displaced manner in the Y direction such that the rack portions **83** overlap with each other as viewed in the Y direction, the installation area for the rack portions **83** as viewed in the Y direction can be reduced.

(162) In addition, in the present embodiment, two pairs of two of the four rack portions **83y** to **83k** are arranged in a displaced manner in the Y direction. That is, it can be said that with respect to in the rotational axis direction of the rotary (Y direction), the range in which the first rack gear is disposed and the range in which the second rack gear is disposed overlap with each other, and the range in which the third rack gear is disposed and the range in which the fourth rack gear is

disposed overlap with each other. In addition, it can be said that with respect to the Y direction, the range in which the first rack gear and the second rack gear are disposed does not overlap with the range in which the third rack gear and the fourth rack gear are disposed. As a result of this, the size of the rotary body **90** in the Y direction can be reduced as compared with a case where all the four rack portions **83y** to **83k** are arranged in a displaced manner in the Y direction.

(163) Moving Elements of Tray

(164) Elements related to the movement of the trays **80y** to **80k** disposed in the rotary body **90** will be described with reference to FIGS. **11A**, **11B**, **12A**, and **12B**. FIGS. **11A** and **11B** are each a perspective view of elements related to the movement of the tray **80k**. FIGS. **12A** and **12B** are each a section view of elements related to the movement of the tray **80k**.

(165) In the present embodiment, the trays **80y** to **80k** are respectively driven by receiving the driving force of the motor **M2** transmitted to the pinion gears **94y** to **94k** via the driving racks **15L** and **15R** serving as transmission devices. Here, the elements for moving the tray **80k** with respect to the rotary body **90** will be described, and since the elements for moving the trays **80y** to **80c** with respect to the rotary body **90** are substantially the same as the elements for moving the tray **80k**, description thereof will be omitted.

(166) FIG. **11A** illustrates a state in which the tray **80k** is inside the rotary body **90**, that is, in a state in which the toner cartridge **70k** is attached to the developing unit **50k**. That is, FIG. **11A** illustrates a state in which the tray **80k** is at the accommodation position, and corresponds to a state in which the toner cartridge **70k** is at the attached position with respect to the developing frame **53k** illustrated in FIG. **4A**. FIG. **11B** illustrates a state in which the tray **80k** has slid to the outside of the rotary body **90**. That is, FIG. **11B** illustrates a state in which the tray **80k** is at the attached position, and corresponds to a state in which the toner cartridge **70k** is at the retracted position with respect to the developing frame **53k** illustrated in FIG. **4B**.

(167) The apparatus body **1A** of the present embodiment includes the driving racks **15L** and **15R** serving as driving gears that drive the pinion gears **94**. The driving racks **15** are each driven by the motor **M2** via the unillustrated drive transmission mechanism.

(168) As described above, two rack portions **83k** are formed at respective end portions of the tray **80k** in the Y direction. Two pinion gears **94k** and the driving racks **15L** and **15R** are disposed at positions respectively corresponding to the rack portions **83k** at the respective end portions. That is, the apparatus body **1A** of the present embodiment includes the driving racks **15L** and **15R** serving as the first driving gear and the second driving gear. It can be said that the driving rack **15L** is an example of a first driving gear, and the driving rack **15R** is an example of a second driving gear. However, this numbering is merely used for the sake of convenience of description, and can be appropriately interchanged in principle. In the case where there is no need to distinguish the driving racks **15L** and **15R** from each other, the driving racks will be each described as a “driving rack **15**”.

(169) The rack portions **83** of the present embodiment are configured as a pair of rack gears, and the pinion gears **94** of the present embodiment are configured as a pair of pinion gears. The pair of rack gears and the pair of pinion gears are disposed on the two end sides (i.e., one end side and the other end side) of the support member (tray **80**) in the Y direction in the present embodiment, but the pair of rack gears and the pair of pinion gears may be disposed at other positions. The rack portions **83k** and the pinion gears **94k** of the moving device **85k** corresponding to the tray **80k** can be respectively referred to as examples of a first pair of rack gears and a first pair of pinion gears. The rack portions **83y** to **83c** and the pinion gears **94y** to **94c** of the moving devices **85y** to **85c** respectively corresponding to the other trays **80y** to **80c** can be respectively referred to as examples of second pairs of rack gears and second pairs of pinion gear.

(170) One of the pair of rack gears engages with one of the pair of pinion gears, and the other of the pair of rack gears engages with the other of the pair of pinion gears. At least one of the pair of pinion gears is driven by the driving rack **15L** serving as a first driving rack. In the present embodiment, two of the pair of pinion gears are simultaneously driven by the driving racks **15L** and

15R serving as a first driving rack and a second driving rack. As a result of this, rotation of the tray **80** is less likely to occur, and thus stable movement of the toner cartridge **70** is made possible.

(171) To be noted, a configuration in which the tray **80** includes one rack portion **83** and is moved by one driving rack **15** via one pinion gear **94** may be employed.

(172) The tray **80k** is held to be slidable in a direction parallel to the guided portion **82k** (that is, the movement direction **Dk**) with respect to the rotary body **90**. The driving rack **15** is held to be slidable in a direction intersecting with the movement direction **Dk** of the tray **80k** with respect to the apparatus body **1A**. The driving rack **15** is configured to slide (reciprocate) in a first direction (vertically upward direction in the present embodiment) and a second direction (vertically downward direction in the present embodiment) opposite to the first direction, with respect to the apparatus body **1A**. That is, the movement direction of the driving rack **15** of the present embodiment is a direction intersecting with (preferably orthogonal to) both the movement direction **Dk** of the tray **80k** and the direction of the rotational axis **90C** of the rotary body **90** (**Y** direction).

(173) A tray moving operation of sliding the tray **80k** between the accommodation position and the detachment position will be described with reference to FIGS. **11A** and **11B**. The tray moving operation of the tray **80k** is performed by the motor **M2** illustrated in FIG. **2**, an unillustrated drive transmission mechanism, the driving rack **15**, the pinion gear **94k**, and the rack portion **83k**.

(174) First, a tray moving operation (tray pull-out operation) of detaching the toner cartridge **70k** from the rotary body **90** will be described. In a state before the tray pull-out operation is started, the driving rack **15** illustrated in FIG. **11A** is positioned below a position where the driving rack **15** engages with the pinion gear **94k**. In addition, as described above, in the replacement operation of the toner cartridge **70k**, the rotary body **90** takes the replacement posture illustrated in FIG. **4B** for the toner cartridge **70k**.

(175) When the tray pull-out operation is started, the driving racks **15** are slid upward with respect to the apparatus body **1A** by the driving force of the motor **M2**. In the course of movement of the driving racks **15**, the driving racks **15** engage with the pinion gears **94k**, and the pinion gears **94k** are rotationally driven.

(176) As illustrated in FIG. **11B**, the pinion gears **94k** are rotationally driven in an arrow direction in FIG. **11B**, and thus the driving force is input to the rack portions **83k** engaged with the pinion gears **94k**. As a result of this, the tray **80k** is pushed to the outside of the apparatus, and moves from the accommodation position to the detachment position with respect to the rotary body **90**. The movement direction of the tray **80k** in this movement is guided to the predetermined movement direction **Dk** by the engagement between the guided portion **82k** and the guide portion **97k** of the rotary body **90** illustrated in FIG. **7A**. As a result of the tray **80k** moving from the accommodation position to the detachment position, the toner cartridge **70k** is moved from the attached position to the retracted position with respect to the developing unit **50k**.

(177) In a state in which the tray **80k** is positioned at the detachment position and the toner cartridge **70k** is positioned at the retracted position, the user can attach and detach the toner cartridge **70k** to and from the tray **80k**.

(178) The tray moving operation (tray pull-in operation or tray insertion operation) at the time of attaching the toner cartridge **70** to the rotary body **90** is performed in a process reversed from the tray pull-out operation. In the state before the tray pull-in operation is started, the driving racks **15** are positioned above positions where the driving racks **15** engage with the pinion gears **94k**. For example, the tray pull-in operation is started in response to the user operating a predetermined operation portion. When the tray pull-in operation is started, the driving racks **15** are slid downward with respect to the apparatus body **1A** by the driving force of the motor **M2**. Here, the rotational direction of the motor **M2** in the tray pull-in operation is opposite to that in the tray pull-out operation. In the course of movement of the driving racks **15**, the driving racks **15** engage with the pinion gears **94k**, and thus the pinion gears **94k** are rotationally driven.

(179) As a result of the pinion gears **94k** being rotationally driven in a direction opposite to the

arrow of FIG. 11B, the driving force is input to the rack portions **83k** engaged with the pinion gears **94k**. As a result of this, the tray **80k** is pulled into the apparatus, and moves from the detachment position to the accommodation position with respect to the rotary body **90**. The movement direction of the tray **80k** is guided to the movement direction Dk opposite to the arrow of FIG. 11B by the engagement between the guided portion **82k** and the guide portion **97k** of the rotary body **90**. As a result of the tray **80k** moving from the detachment position to the accommodation position, the toner cartridge **70k** is moved from the retracted position to the attached position with respect to the developing unit **50k**.

(180) Movement of the tray **80k** and the toner cartridge **70k** for black has been described above, and movement of the other trays **80y** to **80c** and toner cartridges **70y** to **70c** is also performed by a similar mechanism. That is, the driving racks **15** transmit drive to the pinion gears **94y** to **94c** in respective replacement postures of the toner cartridges.

(181) The motor M2 provided in the apparatus body **1A** and the transmission device including the driving racks **15** (**15L** and **15R**) and the drive transmission mechanism constitute the driving device **98** for driving the moving devices **85** provided in the rotary body **90**.

(182) As described above, in the present embodiment, a plurality of moving devices **85k** to **85y** corresponding to the plurality of toner cartridges **70k** to **70y** are disposed in the rotary body **90**. The driving device **98** of the apparatus body **1A** is a common driving device that drives the plurality of moving devices **85k** to **85y** (plurality of driven devices) of the rotary body **90**.

(183) In addition, in the present embodiment, the driving target of the driving device **98** switches in accordance with the rotation of the rotary body **90**. In other words, the driving device of the present embodiment includes the driving racks **15** each serving as a transmission member that transmits the driving force of the drive source. The driving device can take a state in which the transmission member engages with a first driven portion (e.g., pinion gear **94k**) in a manner capable of transmitting the driving force and a state in which the transmission member engages with a second driven portion (e.g., pinion gear **94m**) in a manner capable of transmitting the driving force. In addition, the driving device can take a state in which the transmission member is disengaged from the first driven portion and the second driven portion.

(184) As described above, the pinion gears **94y** to **94k** are held by the rotary body **90**. Therefore, when the rotary body **90** rotates, it is preferable that the engagement between the driving racks **15** and the pinion gears **94y** to **94k** is released.

(185) FIG. 12A illustrates a state in which the tray **80k** is inside the rotary body **90**, that is, a state in which the tray **80k** is at the accommodation position. FIG. 12B illustrates a state in which the tray **80k** has moved to the outside of the rotary body **90**, that is, a state in which the tray **80k** has moved to the detachment position.

(186) As illustrated in FIG. 12A, when the tray **80k** is inside the rotary body **90**, the driving racks **15** are positioned in a lower portion in the apparatus body **1A**. At this time, the driving racks **15** are retracted from the pinion gears **94k**. Therefore, the rotary body **90** can be rotated without being interrupted by the driving racks **15**. More specifically, the driving racks **15** can be retracted to the outside of the rotational trajectory of the rotary body **90** indicated by dotted lines in FIGS. 12A and 12B.

(187) As described above, by rotationally driving the motor M2 in a normal direction and a reverse direction, the tray **80** attached to the rotary body **90** can be moved from the accommodation position to the detachment position and from the detachment position to the accommodation position with respect to the rotary body **90**. That is, the driving device of the present embodiment can drive the moving devices such that not only the toner cartridges move from the attached position to the retracted position but also the toner cartridges move from the retracted position to the attached position.

(188) Here, as described above, in the present embodiment, the movement amount of the tray **80** in toner cartridge replacement is changed in accordance with the size of the toner cartridge **70**.

Specifically, the movement distance L1 of the movement of the black tray **80k** from the accommodation position to the detachment position is larger than the movement distance L2 of the movement of the other trays **80y** to **80c** from the accommodation position to the detachment position as illustrated in FIGS. 7A and 7B.

(189) Therefore, in the present embodiment, when moving the toner cartridges **70y** to **70k** from the attached position to the retracted position, a value obtained by dividing the speed of the rack portions **83k** by the speed of the driving racks **15** is larger than a value obtained by dividing the speed of the rack portions **83y** to **83c** by the speed of the driving racks **15**.

(190) For example, as illustrated in FIG. 10, the pinion gear **94y** is formed as a stepped gear including a large-diameter gear **941** that engages with the driving rack **15** and a small-diameter gear **942** that has a smaller pitch radius than the large-diameter gear **941** and engages with the rack portion **83y**. The pinion gears **94m** and **94c** are formed as similar stepped gears. In contrast, the pinion gear **94k** is formed such that a portion thereof that engages with the driving rack **15** and a portion thereof that engages with the rack portion **83k** have the same pitch radius. In this case, the pitch radius of the pinion gear **94k** can be equal to the pitch radius of the large-diameter gear **941** of the pinion gears **94y** to **94c**. According to this configuration, the movement distance of the rack portions **83k** can be made larger than the movement distance of the other rack portions **83y** to **83c** even in the case where the movement distance of the driving racks **15** is the same. That is, the movement distance L1 of the movement of the black tray **80k** from the accommodation position to the detachment position can be made larger than the movement distance L2 of the movement of the other trays **80y** to **80c** from the accommodation position to the detachment position.

(191) In addition, by forming the pinion gears **94y** to **94c** as stepped gears, the movement distance L1 of the tray **80k** can be made larger than the movement distance L2 of the other trays **80y** to **80c** even in the configuration in which the pinion gears **94y** to **94k** receive the driving force from the same driving racks **15**.

(192) To be noted, the pinion gear **94k** may be formed as a stepped gear instead of (or in combination with) the configuration in which the pinion gears **94y** to **94c** are formed as stepped gears. In this case, a portion of the pinion gear **94k** that engages with the driving rack **15** may be formed as a small-diameter gear, and a portion of the pinion gear **94k** that engages with the rack portion **83k** may be formed as a large-diameter gear having a larger pitch radius than the small-diameter gear. In addition, the stepped gear is an example of a speed reduction mechanism, and may be replaced by a known speed reduction mechanism that makes the movement amount of a member on the input side (drive source side) smaller than a movement amount of a member on the output side (tray **80** side).

(193) In addition, the movement amount of the driving rack **15** when the toner cartridge **70k** moves from the attached position to the retracted position may be made larger than the movement amount of the driving rack **15** when the toner cartridges **70y** to **70c** move from the attached position to the retracted position.

(194) Incidentally, in the case where the distance of the movement of the toner cartridge **70** from the attached position to the retracted position is smaller, the movement time of the toner cartridge **70** can be made shorter, and the time in which the user waits for the toner cartridge **70** to move can be made shorter. If a configuration in which the movement amount of the driving racks **15** for the toner cartridge **70k** is larger than the movement amount of the driving racks **15** for the toner cartridges **70y** to **70c** is employed as described above, the time in which the user waits for the toner cartridges **70y** to **70c** to move can be made shorter.

(195) According to the configurations described above, the movement distance L1 can be made larger than the movement distance L2. These configurations may be employed in combination.

Modification Example

(196) Although a configuration in which the driven portion includes the pinion gears **94** that engage with both the driving racks **15** and the rack portions **83** has been described, the driven portion may

include a gear that engages with the driving rack **15** and a gear that engages with the rack portion **83**.

(197) In addition, the configuration of the moving device **85** that move the tray **80** is not limited to a so-called rack-and-pinion configuration. For example, the member corresponding to the pinion gear **94** may be replaced by a roller that receives the drive of the motor M2 to rotate, and the tray **80** may be moved by friction between the roller and the tray **80**.

(198) In addition, in the case of using the roller that receives the drive of the motor M2 to rotate, the roller may be brought into contact with the toner cartridge **70**. In this case, the toner cartridges **70y** to **70k** can be made directly attachable to and detachable from the rotary body **90** without using the trays **80y** to **80k**. In this case, the moving device **85** is constituted by a roller.

(199) A moving device **85'** serving as a modification example will be described with reference to FIGS. **20A** and **20B**. FIGS. **20A** and **20B** are diagrams illustrating the moving device **85'** according to the present modification example. The moving device **85'** includes a rotary member **494a** that receives the driving force of the motor M2 to rotate.

(200) In the present modification example, the direction of the rotational axis of the rotary member **494a** is parallel to the direction of the rotational axis **90C** of the rotary body **90**. The rotary member **494a** abuts the toner cartridge **70** and rotates, and thus the toner cartridge **70** can reciprocate between the attached position indicated by a solid line in FIG. **20B** and the retracted position indicated by a dot line in FIG. **20B**.

(201) To be noted, the rotary member **494a** may be a roller that moves the toner cartridge **70** by friction by abutting the toner cartridge **70** and rotating. In addition, the toner cartridge **70** may be moved by a configuration in which the rotary member **494a** is a gear engaging with a gear shape (rack shape) formed on the toner cartridge **70**.

(202) The moving device **85'** may include a plurality of rotary members **494a**. The plurality of rotary members **494a** may be arranged arbitrarily. For example, as illustrated in FIG. **20A**, the moving device **85'** may include a rotary member **494a** that abuts one end portion of the toner cartridge **70** and a rotary member **494a** that abuts the other end portion of the toner cartridge **70** in the longitudinal direction of the toner cartridge **70** parallel to the rotational axis **90C**. In addition, the moving device **85'** may include a rotary member **494a** that abuts the center of the toner cartridge **70**.

(203) In addition, the moving device **85'** may include only one rotary member **494a**. In this case, the moving device **85'** may be disposed at an arbitrary position. For example, the moving device **85'** may include a rotary member **494a** that abuts the center of the toner cartridge **70**.

(204) Further, the rotary member **494a** may be urged toward the toner cartridge **70**. In addition, the moving device **85'** may include a driven roller **494b** as illustrated in FIG. **20B**. The toner cartridge **70** is interposed between the rotary member **494a** and the driven roller **494b**. To be noted, with respect to the longitudinal direction of the toner cartridge **70**, the position of the rotary member **494a** and the position of the driven roller **494b** may overlap with each other, or may be different from each other. In addition, at least one of the rotary member **494a** and the driven roller **494b** may be urged toward the toner cartridge **70**.

(205) In addition, the rotary member **494a** and the driven roller **494b** may be provided in the rotary body **90**.

(206) In the first embodiment, as an embodiment in which the door **14** moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **70** from the attached position to the retracted position, a configuration in which the tray **80** pushes the door **14** has been described. Instead of this configuration, for example, a configuration in which part of the toner cartridge **70** pushes the door **14** when the toner cartridge **70** moves from the attached position to the retracted position, and thus the door **14** moves from the closed position to the open position, may be employed. In addition, the door **14** may be moved by transmitting a driving force from a member of the moving device **85** different from the tray **80** (support member)

or part of the driving device **98**. For example, an input gear may be attached to the door **14**, and rotation may be transmitted from the pinion gear **94** of the moving device **85** illustrated in FIG. **11A** or the driving rack **15** of the driving device **98** to this input gear via a gear train.

(207) Further, the embodiment in which the door **14** moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **70** from the attached position to the retracted position is not limited to mechanical interconnection, and the movement of the toner cartridge **70** may be interconnected with the movement of the door **14** by electrical control. For example, the door **14** is configured to be driven by a drive source (for example, a solenoid) different from the driving device **98** for moving the toner cartridge **70**.

Further, control of the driving device **98** to move the toner cartridge **70** from the attached position to the retracted position (control of the motor **M2**) and control of the other drive source described above to move the door **14** from the closed position to the open position are executed in the same sequence process.

(208) The sequence process is a series of processing performed by the controller described above to enable replacement of the toner cartridge **70** by the user and then restoring the image forming apparatus **1** to a state in which the image forming operation can be executed again in the case where an instruction to perform the toner cartridge replacement operation is received from the user. A period in which the driving device **98** is driven to move the toner cartridge **70** from the attached position to the retracted position and a period in which the other drive source is driven to move the door **14** from the closed position to the open position may overlap (progress in parallel) with each other at least partially.

Second Embodiment

(209) A second embodiment will be described with reference to FIGS. **13** to **14C**. Although the pinion gear **94** is driven by the driving rack **15** in the first embodiment, the pinion gear **94** may be driven by a different mechanism. In the present embodiment, a configuration employing a swing gear will be described as an example. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

(210) FIGS. **13** to **14C** are schematic diagrams for describing a configuration for moving the tray **80** in the second embodiment. An apparatus body **101A** of an image forming apparatus **101** according to the second embodiment includes, as a driving device, the motor **M2** serving as a drive source, and a swing gear unit **115** serving as a transmission device.

(211) A moving device (driven device) that is driven by the driving device to move the tray **80** is disposed in the rotary body **90**. That is, the rotary unit **90U** includes the rotary body **90** and the moving device including the tray **80**.

(212) In the second embodiment, a plurality of moving devices are disposed in correspondence with the plurality of toner cartridges **70y** to **70k**. The moving devices each include a driven portion **194** (one of driven portions **194y** to **194k**), and a rack portion **83** (one of rack portions **83y** to **83k**). The swing gear unit **115** transmits the driving force of the motor **M2** to the driven portion **194**, and moves the tray **80** via the rack portion **83**.

(213) Similarly to the first embodiment in which the pinion gears **94y** to **94k** are driven by the driving racks **15**, the first to fourth driven portions **194y** to **194k** are each driven by the swing gear unit **115**. The first to fourth driven portions **194y** to **194k** are respectively engaged with the rack portions **83y** to **83k**, and respectively move the trays **80y** to **80k**.

(214) The configuration in which each of the first to fourth driven portions **194y** to **194k** is driven by the swing gear unit **115** is substantially the same. In addition, the configuration for moving each of the trays **80y** to **80k** via each of the first to fourth driven portions **194y** to **194k** and each of the rack portions **83y** to **83k** is the same. Therefore, in the present embodiment, the first to fourth driven portions **194y** to **194k**, the rack portions **83y** to **83k**, and the trays **80y** to **80k** will be

respectively referred to as driven portions **194**, rack portions **83**, and trays **80** without distinguishing each other.

(215) As illustrated in FIG. **14A**, the swing gear unit **115** includes an input gear **115a**, an output gear **115b**, and an arm **115c**. The input gear **115a** is rotationally driven by the motor **M2**. The arm **115c** rotatably supports the output gear **115b** (such that the output gear **115b** can rotate) and allows the output gear **115b** to move (swing or revolve) around the input gear **115a**.

(216) Further, the swing gear unit **115** includes an arm restriction member **115d** that restricts the arm **115c**. The arm restriction member **115d** includes a restriction portion **115d2** and a restriction rack portion **115d1**. The restriction rack portion **115d1** is engaged with the input gear **115a**.

(217) In addition, the driven portion **194** includes a driven gear **194a** and a pinion gear **194b**. The pinion gear **194b** is engaged with the rack portion **83** of the tray **80**.

(218) In the case of moving the tray **80** from the accommodation position to the detachment position, that is, in the case of moving the toner cartridge **70** from the attached position to the retracted position, the motor **M2** rotationally drives the input gear **115a** in a clockwise direction in FIGS. **14A** to **14C**.

(219) When the input gear **115a** rotates in the clockwise direction in the drawing, the arm **115c** swings in the clockwise direction in the drawing similarly to the input gear **115a** by a frictional force received from the input gear **115a**. The output gear **115b** supported by the arm **115c** rotates in a clockwise direction about the input gear **115a** together with the arm **115c**, and engages with the driven gear **194a** as illustrated in FIG. **14B**. At this time, the reaction force that the output gear **115b** receives from the driven gear **194a** preferably acts in such a direction as to move the output gear **115b** and the arm **115c** in a clockwise direction about the input gear **115a**. That is, the direction of the reaction force that the output gear **115b** receives from the driven gear **194a** when rotating in a counterclockwise direction in FIG. **14B** while pressing the driven gear **194a** is preferably such a direction that a moment in a clockwise direction in the drawings with respect to the rotational axis of the input gear **115a** is generated. As a result of this, the engagement between the output gear **115b** and the driven gear **194a** is more likely to be maintained.

(220) As illustrated in FIG. **14C**, when the input gear **115a** rotates in the clockwise direction in a state in which the output gear **115b** is engaged with the driven gear **194a**, the driven gear **194a** drives the pinion gear **194b**. As a result of the rack portion **83** being moved by the rotation of the pinion gear **194b**, the tray **80** moves from the accommodation position to the detachment position. Further, the toner cartridge **70** moves from the attached position to the retracted position.

(221) To be noted, the positional relationship between the toner cartridge **70** and the frame **16** of the apparatus body **101A** when the toner cartridge **70** is at the retracted position and the positional relationship between the toner cartridge **70** and the rotary body **90** are the same as those described in the first embodiment, and therefore description thereof will be omitted.

(222) Meanwhile, in the course of clockwise rotation of the input gear **115a** described above, the arm restriction member **115d** including the restriction rack portion **115d1** that engages with the input gear **115a** moves down. Then, the arm restriction member **115d** is positioned at such a position that the restriction portion **115d2** restricts the arm **115c**. The restriction portion **115d2** restricts such movement of the arm **115c** that the output gear **115b** disengages from the driven gear **194a**.

(223) In the case of moving the tray **80** from the detachment position to the accommodation position, that is, in the case of moving the toner cartridge **70** from the retracted position to the attached position, the motor **M2** rotationally drives the input gear **115a** in a counterclockwise direction in FIGS. **14A** to **14C**.

(224) When the input gear **115a** rotates in the counterclockwise direction, the output gear **115b** supported by the arm **115c** receives a force in such a direction as to move in the counterclockwise direction about the input gear **115a** together with the arm **115c**. However, since the movement of the arm **115c** is restricted by the arm restriction member **115d**, the engagement between the output

gear **115b** and the driven gear **194a** is maintained.

(225) When the input gear **115a** rotates in the counterclockwise direction in a state in which the output gear **115b** is engaged with the driven gear **194a**, the pinion gear **194b** is driven by the driven gear **194a**, and the tray **80** moves from the detachment position to the accommodation position. Then, the toner cartridge **70** moves from the retracted position to the attached position.

(226) After the tray **80** has moved to the accommodation position, the restriction of the atm **115c** by the restriction portion **115d2** of the arm restriction member **115d** is cancelled. As a result, the rotation of the input gear **115a** in the counterclockwise direction moves the output gear **115b** in the counterclockwise direction about the input gear **115a** together with the arm **115c**, and thus the engagement between the output gear **115b** and the driven gear **194a** is cancelled.

(227) According to the configuration described above, the tray **80** can be moved from the accommodation position to the detachment position. In addition, the toner cartridge **70** can be moved from the attached position to the retracted position. Further, the tray **80** can be moved from the detachment position to the accommodation position. In addition, the toner cartridge **70** can be moved from the retracted position to the attached position.

Modification Example

(228) Although the driving force of the motor **M2** is transmitted to the tray **80** via the driven gear **194a** and the pinion gear **194b** in the present embodiment, one gear may be configured to engage with the output gear **115b** and the rack portion **83**.

(229) In addition, similarly to the first embodiment, the configuration for moving the tray **80** is not limited to a so-called rack-and-pinion configuration.

(230) As yet another modification example, a partially-toothed gear may be used as a drive transmission mechanism from the motor **M2** to the pinion gear engaged with the rack portion **83** instead of the configuration of the first and second embodiments. For example, a partially-toothed gear that engages with a pinion gear of the rotary body **90** may be provided in the apparatus body **1A**, and the partially-toothed gear may be rotationally driven by the motor **M2**. In this case, the partially-toothed gear engages with a pinion gear that engages with the rack portion **83** of the tray **80** facing the opening **16a** when the rotary body **90** takes the replacement posture illustrated in FIG. **4B**. In the black replacement posture illustrated in FIG. **12A**, the partially-toothed gear engages with the pinion gear **94k**. By switching the rotation direction of the partially-toothed gear, the movement direction of the tray **80** can be switched. In addition, it is preferable that the engagement between the partially-toothed gear and the pinion gear **94** is cancelled in a period other than when the tray **80** is moved with respect to the rotary body **90**.

Third Embodiment

(231) A third embodiment will be described with reference to FIGS. **15** to **16B**. Although the toner cartridge **70** is moved from the attached position to the retracted position by the driving force of the motor **M2** of the apparatus body **1A** in the first and second embodiments, the configuration for moving the toner cartridge **70** is not limited to this. In the present embodiment, a configuration in which the toner cartridge **70** is moved by an elastic force (urging force) of a spring (elastic member) will be described as an example. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

(232) As illustrated in FIG. **15**, in the image forming apparatus according to the present embodiment, the toner cartridge **70** is moved from the attached position to the retracted position by using an urging member **294** serving as an example of a moving device. More specifically, the rotary body **90** of the image forming apparatus includes an urging member **294k** corresponding to the toner cartridge **70k** and the tray **80k**, and urging members **294y** to **294c** corresponding to the toner cartridges **70y** to **70c** and the trays **80y** to **80c**. That is, the rotary of the present embodiment includes first to fourth urging members corresponding to the first to fourth toner cartridges. The

urging member **294k** is an example of the first urging member. The urging members **294y** to **294c** are each an example of a second urging member. To be noted, this numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

(233) The urging member **294** of the present embodiment includes an elastic member, and is attached to the rotary body **90**. In the present embodiment, a compression spring can be used as the elastic member.

(234) In the present embodiment, the urging member **294** urges the tray **80**, and thus the tray **80** moves from the accommodation position to the detachment position and the toner cartridge **70** moves from the attached position to the retracted position. Although the urging member **294** urges the toner cartridge **70** via the tray **80** in the present embodiment, the urging member **294** may directly urge the toner cartridge **70**.

(235) That is, although the urging member **294** of the present embodiment is configured to be into contact with the tray **80**, the urging member **294** may be in contact with the toner cartridge **70**. In this case, the tray **80** may be omitted.

(236) To be noted, also in the present embodiment, the toner cartridge **70k** and the tray **80k** are larger in size than the other toner cartridges **70y** to **70c** and the trays **80y** to **80c**. Therefore, the distance by which the toner cartridge **70k** moves from the attached position to the retracted position is larger than the distance by which the toner cartridges **70y** to **70c** move from the attached position to the retracted position. In addition, the distance by which the tray **80k** moves from the accommodation position to the detachment position is larger than the distance by which the trays **80y** to **80c** move from the accommodation position to the detachment position.

(237) In a state in which the tray **80k** is at the accommodation position and the toner cartridge **70k** is at the attached position, the elastic member of the urging member **294k** is deformed. Similarly, in a state in which the trays **80y** to **80c** are at the accommodation position and the toner cartridges **70y** to **70c** are at the attached position, the elastic members of the urging members **294y** to **294c** are deformed.

(238) In this state, the urging force of the elastic member of the urging member **294k** may be set to be larger than the urging force of the elastic members of the urging members **294y** to **294c**. In addition, the deformation amount of the elastic member of the urging member **294k** may be set to be larger than the deformation amounts of the elastic members of the urging members **294y** to **294c**. As a result of this, the distance by which the toner cartridge **70k** moves from the attached position to the retracted position can be made larger than the distance by which the toner cartridges **70y** to **70c** move from the attached position to the retracted position. In addition, the distance by which the tray **80k** moves from the accommodation position to the detachment position can be made larger than the distance by which the trays **80y** to **80c** move from the accommodation position to the detachment position.

(239) To be noted, when the rotary body **90** rotates, it is preferable that the movement of the toner cartridge **70** to the retracted position and the movement of the tray **80** to the detachment position are restricted.

(240) Therefore, as illustrated in FIG. **16A**, the movement of the tray **80** to the detachment position and the movement of the toner cartridge **70** to the retracted position may be restricted by a restriction member **295**. The restriction member **295** is movable to a restricting position where the restriction member **295** restricts movement of the tray **80** to the detachment position and movement of the toner cartridge **70** to the retracted position, and to a releasing position where the restriction member **295** allows movement of the tray **80** to the detachment position and movement of the toner cartridge **70** to the retracted position.

(241) In the present embodiment, the restriction member **295** is urged toward the restricting position. In a state in which the rotary body **90** is in the replacement posture, when the restriction member **295** is moved to the releasing position, the urging member **294** moves the tray **80** to the detachment position, and thus the toner cartridge **70** moves to the retracted position. The movement

of the restriction member **295** to the releasing position may be manually performed by the user, and the apparatus body **1A** may include a releasing means or a releasing mechanism that moves the restriction member **295**. As the releasing means (releasing mechanism), a drive source such as a solenoid or a motor can be used. To be noted, one of the motors **M1**, **M2**, and **M3** may be used as the drive source.

(242) In addition, the urging member **294** may be configured to switch the urging state each time the toner cartridge **70** or the tray **80** is pushed in.

(243) For example, as illustrated in FIG. **16B**, the image forming apparatus may include a received portion **296** interlocked with the urging member **294**, and a receiving portion **297** that receives the received portion **296**. For example, the received portion **296** is provided on the tray **80**, and the receiving portion **297** is provided on the rotary body **90**. The received portion **296** is urged rightward in the drawing by the urging force of the urging member **294**.

(244) In a state in which the received portion **296** is at a position PP, the urging force of the urging member **294** is received by a first recess portion **297d** of the receiving portion **297**, and the received portion **296** stays in the position PP. In this state, the tray **80** is positioned at the detachment position, and the toner cartridge **70** is positioned at the retracted position.

(245) When the tray **80** is pushed in toward the accommodation position or the toner cartridge **70** is pushed in toward the attached position against the urging force of the urging member **294** in this state, the received portion **296** moves to a position Pt1 along a first guiding surface **297a** of the receiving portion **297**. Then, when the force to push in the tray **80** or the toner cartridge **70** is no longer present, the urging force of the urging member **294** moves the received portion **296** to a position PR along the receiving portion **297**. At this time, the urging force of the urging member **294** is received by a second recess portion **297e** of the receiving portion **297**. As a result, the tray **80** is positioned at the accommodation position, and the toner cartridge **70** is positioned at the attached position.

(246) When the tray **80** or the toner cartridge **70** is pushed in again against the urging force of the urging member **294** in this state, the received portion **296** moves to a position Pt2 along a second guiding surface **297b** of the receiving portion **297**. Then, when the force to push in the tray **80** or the toner cartridge **70** is no longer present, the urging force of the urging member **294** moves the received portion **296** to the position PP along a third guiding surface **297c** of the receiving portion **297**. As a result, the tray **80** is positioned at the detachment position, and the toner cartridge **70** is positioned at the retracted position.

(247) To be noted, in the case where the receiving portion **297** is attached to the rotary body **90**, the urging member **294** may include the elastic member and the received portion **296**. In this case, in a state in which the received portion **296** is positioned at the position PR, the urging force of the urging member **294** is received by the second recess portion **297e** of the receiving portion **297**, and does not act on the tray **80** or the toner cartridge **70**. In addition, when the received portion **296** moves from the position Pt2 to the position PP, the urging force of the urging member **294** acts on the tray **80** and the toner cartridge **70**, the tray **80** is positioned at the detachment position, and the toner cartridge **70** is positioned at the retracted position. To be noted, a member corresponding to the received portion **296** may be provided in the rotary body **90**, and a member corresponding to the receiving portion **297** may be provided in the urging member **294** or the tray **80**.

(248) To be noted, the image forming apparatus may include a locking portion **298** (tray holding member) illustrated in FIG. **16C**. Although the locking portion **298** engages with the tray **80** in the present embodiment, the locking portion **298** may engage with the toner cartridge **70**. The locking portion **298** holds the tray **80** at the accommodation position or holds the toner cartridge **70** at the attached position in a state in which the urging member **294** is not urging the tray **80** or the toner cartridge **70**. For example, also in a state in which the direction of the gravity acting on the toner cartridge **70** has come to match the retraction direction or has become close to the retraction direction of the toner cartridge **70** due to the rotation of the rotary body **90**, the toner cartridge **70** is

held at the attached position by the locking portion **298**. When the urging force of the urging member **294** acts on the tray **80** or the toner cartridge **70**, the locking portion **298** is moved by the tray **80** or the toner cartridge **70**. To be noted, the position where the locking portion **298** engages with a locked portion of the tray **80** or the toner cartridge **70** is not limited to the position of FIG. **16C**.

(249) As the locking portion **298**, for example, the same part as the projection portion **95** provided in the rotary body **90** of the first embodiment illustrated in FIGS. **7A** and **7B** can be used.

Modification Example

(250) Although the elastic member is a compression spring in the present embodiment, the elastic member may be a torsion coil spring, a plate spring, or a tension spring each provided at an appropriate position.

(251) In addition, for example, magnets may be used as the urging member. For example, a first magnet may be attached to the rotary body **90**, and the tray **80** may be urged toward the retracted position by a repulsive force between the first magnet and a second magnet attached to the toner cartridge **70** or the tray **80**.

Fourth Embodiment

(252) A fourth embodiment will be described with reference to FIGS. **17A** and **17B**. Although the toner cartridge **70** is moved from the attached position to the retracted position by the driving force of the motor **M2** of the apparatus body **1A** in the first and second embodiments and by the urging member **294** in the third embodiment, the configuration for moving the toner cartridge **70** is not limited to these. In the present embodiment, a configuration in which the toner cartridge **70** is moved by using a solenoid will be described as an example. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

(253) FIGS. **17A** and **17B** are each a diagram for describing a configuration for moving the tray **80** in the fourth embodiment. In the image forming apparatus according to the present embodiment, the toner cartridge **70** is moved from the attached position to the retracted position by using a solenoid unit **394** serving as an example of a moving device. More specifically, the rotary body **90** of the present embodiment includes a solenoid unit **394k** corresponding to the toner cartridge **70k** and the tray **80k**. In addition, the rotary body **90** of the present embodiment includes solenoid units **394y** to **394c** corresponding to the toner cartridges **70y** to **70c** and the trays **80y** to **80c**.

(254) The solenoid unit **394** includes a coil and a movable member, and the movable member is moved by a magnetic field induced in the coil. The movable member is capable of moving such that the tray **80** moves from the accommodation position to the detachment position or the toner cartridge **70** moves from the attached position to the retracted position.

(255) In addition, the movable member is capable of moving such that the tray **80** moves from the detachment position to the accommodation position or the toner cartridge **70** moves from the retracted position to the attached position.

(256) To be noted, as illustrated in FIG. **17B**, the image forming apparatus may include a locking portion **397** (tray holding member). Although the locking portion **397** engages with the tray **80** in the present embodiment, the locking portion **397** may engage with the toner cartridge **70**. The locking portion **397** holds the tray **80** at the accommodation position or holds the toner cartridge **70** at the attached position in a state in which the movable member of the solenoid unit **394** is not urging the tray **80** or the toner cartridge **70**. That is, the locking portion **397** restricts movement of the tray **80** and the toner cartridge **70** in a state in which the movable member of the solenoid unit **394** is not urging the tray **80** and the toner cartridge **70**. In contrast, when the movable member of the solenoid unit **394** acts on the tray **80** or the toner cartridge **70**, the locking portion **397** is moved by the tray **80** or the toner cartridge **70**. To be noted, the position where the locking portion **397** engages with a locked portion of the tray **80** or the toner cartridge **70** is not limited to the position

of FIG. 17B.

(257) As the locking portion **397**, for example, the same part as the projection portion **95** provided in the rotary body **90** of the first embodiment illustrated in FIGS. 7A and 7B can be used.

Fifth Embodiment

(258) An image forming apparatus **201** according to a fifth embodiment will be described with reference to FIG. 18. In the present embodiment, a second surface of a body frame is provided at a more inward position than a first surface of the body frame constituting the exterior surface of the image forming apparatus **201**, and the opening/closing member is disposed on the second surface. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

(259) As illustrated in FIG. 18, a frame (body frame) **215** of the image forming apparatus **201** of the present embodiment includes an outer frame **217** and an inner frame **216** accommodated inside the outer frame **217**. The outer frame **217** constitutes the exterior surface (appearance surface) of the image forming apparatus **201**.

(260) A door **214** serving as an opening/closing member of the present embodiment is disposed on a side surface **216b** on the +X side of the inner frame **216**. The door **214** is movable between a closed position where the door **214** covers an opening **216a** provided in the side surface **216b** and an open position where the door **214** allows attachment and detachment of the toner cartridge **70** to and from the rotary body **90** via the opening **216a**, with respect to the inner frame **216**.

(261) The side surface **216b** of the inner frame **216** is positioned more on the inside than a side surface **217a** of the outer frame **217** with respect to the image forming apparatus **201**. The side surface **217a** of the outer frame **217** constitutes at least part of the exterior surface on the +X side of the image forming apparatus **201**. That is, the side surface **217a** is an example of a first surface constituting part of the exterior surface of the image forming apparatus, and the side surface **216b** of the inner frame **216** is an example of a second surface provided at a more inward position than the first surface with respect to the image forming apparatus.

(262) At least part of the side surface **217a** of the outer frame **217** is configured as an openable portion **217b** capable of pivoting (or opening and closing) with respect to the other part of the outer frame **217**. Further, the user is allowed to replace the toner cartridge **70** by accessing a space between the outer frame **217** and the inner frame **216** by, for example, manually opening the openable portion **217b**.

(263) In the present embodiment, for example, after the user has opened the openable portion **217b** of the outer frame **217**, the door **214** moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **70** from the attached position to the retracted position. Also according to such a configuration, the user does not have to open the door **214** manually, which makes it easier for the user to replace the toner cartridge.

(264) To be noted, the side surface **216b** of the inner frame **216** and the side surface **217a** of the outer frame **217** are each a surface extending in a direction intersecting with the horizontal direction (that is, a direction intersecting with the X direction) as viewed in the rotational axis direction of the rotary body **90** (Y direction). The side surface **217a** of the outer frame **217** constitutes part of the exterior surface on the downstream side of the image forming apparatus **201** in a discharge direction in which the sheet S on which an image has been formed is discharged through a discharge port of the apparatus body **1A**. In addition, it can be said that the side surface **217a** of the outer frame **217** constitutes part of the exterior surface on the front side of the image forming apparatus **201**.

(265) Here, in the first embodiment, a configuration in which the sheet S stored in the sheet storage portion **300** projects to the outside of the image forming apparatus **1** with respect to the side surface **16b** of the frame **16** in which the opening **16a** and the door **14** are provided as illustrated in FIG. 1

has been described. In the present embodiment, the outer frame **217** is formed in accordance with the size of a cassette **305** serving as a sheet storage portion that stores the sheet S. Specifically, a side surface **305a** on the downstream side of the cassette **305** in a draw-out direction **Dp** of the cassette **305** constitutes the exterior surface of the image forming apparatus **201** together with the side surface **217a** of the outer frame **217**. As a result of this, the aesthetic appearance of the apparatus can be improved by not projecting the sheet S from the body frame **215**, and thus, for example, a possibility that the user erroneously staining the sheet S can be reduced.

(266) To be noted, the draw-out direction **Dp** is a movement direction in which the cassette **305** is drawn out from a stored position where the cassette **305** is stored in the body frame **215** to a drawn-out position where at least part of the cassette **305** projects to the outside of the body frame **215**. A configuration in which the side surface **305a** of the cassette **305** and the side surface **217a** of the outer frame **217** are formed along the same virtual plane, that is, are flush may be employed.

Sixth Embodiment

(267) An image forming apparatus **301** according to a sixth embodiment will be described with reference to FIGS. **19A** and **19B**. In the present embodiment, a second opening/closing member is movable to a position where the second opening/closing member covers the opening/closing member and a position where the second opening/closing member exposes the opening/closing member. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

(268) As illustrated in FIGS. **19A** and **19B**, the image forming apparatus **301** according to the present embodiment includes an outer door **302**. The outer door **302** is movable between a position illustrated in FIG. **19A** where the outer door **302** covers the door **14** (inner door) at the closed position and a position illustrated in FIG. **19B** where the outer door **302** exposes the door **14**, with respect to the frame **16**. In the state of FIG. **19B**, movement of the door **14** from the closed position to the open position, and attachment and detachment of the toner cartridge **70** to and from the rotary body **90** via the opening **16a** are possible. The outer door **302** is an example of a second opening/closing member movable between a second closed position where the second opening/closing member covers the opening/closing member (door **14**, first opening/closing member) at the closed position and a second open position where the second opening/closing member exposes the opening/closing member so as to allow movement of the opening/closing member from the closed position to the open position, with respect to the body frame.

(269) The outer door **302** is pivotably supported by the frame **16** about a support shaft **302a** provided at a lower end portion of the outer door **302**. As a result of this, the outer door **302** is pivotable with respect to the frame **16** about an axis substantially parallel to the rotational axis direction of the rotary body **90** (Y direction). To be noted, the configuration for making the outer door **302** movable is not limited to this, and for example, a configuration in which the outer door **302** pivots about the rotational axis extending in a direction (Z direction) intersecting with the Y direction as a result of a support shaft provided at a side end portion of the outer door **302** in the Y direction may be employed.

(270) In the present embodiment, for example, after the user has opened the outer door **302** manually, the door **14** moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **70** from the attached position to the retracted position. Also according to such a configuration, the user does not have to open the door **14** manually, which makes it easier for the user to replace the toner cartridge.

Modification Example

(271) To be noted, in the fifth and sixth embodiments, the door **14** may be omitted. In this case, the openable portion **217b** of the outer frame **217** of the fifth embodiment or the outer door **302** of the sixth embodiment may be opened in an interconnected manner with the movement of the toner

cartridge **70** from the attached position to the retracted position. In this case, the user does not have to open the openable portion **217b** or the outer door **302** manually, which makes it easier for the user to replace the toner cartridge similarly to the other embodiments.

Seventh Embodiment

(272) A seventh embodiment will be described with reference to FIG. **21**. In the first to fourth embodiments, a configuration in which the rotary body **90** includes four developing units **50y** to **50k** and a color image can be formed by using toners of four colors has been described. In the present embodiment, a configuration in which a toner cartridge is moved with respect to a developing device capable of forming a monochromatic image will be described. In the description below, it is assumed that elements denoted by the same reference signs as in the first to fourth embodiments have substantially the same configurations and functions as those described in the first to fourth embodiments unless otherwise described, and parts different from the first to fourth embodiments will be mainly described.

(273) As illustrated in FIG. **21**, an image forming apparatus **501** includes a toner cartridge **570** attachable to and detachable from the apparatus body **1A**. In addition, the apparatus body **1A** includes a developing device (developing unit) **590**.

(274) The developing device **590** is an example of a developing means or a developing portion that develops (visualizes) an electrostatic latent image formed on the photosensitive drum **2** into a toner image by using toner. The developing device **590** of the present embodiment develops the electrostatic latent image formed on the photosensitive drum **2** by using black toner.

(275) The developing device **590** includes a developing roller **51**, a supply roller **52**, and a developing blade. The toner cartridge **570** is attached to the developing device **590**. The toner cartridge **570** stores therein black toner to be supplied to the developing device **590**.

(276) The toner cartridge **570** includes a toner frame **571**. The toner frame **571** includes a toner storage portion **571a** that stores toner, and a discharge opening **571b** communicating with the toner storage portion **571a**.

(277) The developing device **590** includes a developing frame (storage frame) **553** including a developing-side storage portion **553a** that stores toner. In addition, the developing frame **553** has an inlet opening **553b** communicating with the developing-side storage portion (toner supply chamber) **553a**.

(278) The toner cartridge **570** is attachable to and detachable from the developing device **590** through the opening **16a** provided in the frame **16** of the apparatus body **1A**. More specifically, the toner cartridge **570** is movable to an attached position and a retracted position where the toner cartridge **570** is retracted from the attached position, through the opening **16a** with respect to the developing frame **553**. In a state in which the toner cartridge **570** is at the attached position with respect to the developing frame **553**, the discharge opening **571b** faces the inlet opening **553b**. That is, the toner storage portion **571a** of the toner cartridge **570** and the developing-side storage portion **553a** of the developing device **590** communicate with each other via the discharge opening **571b** and the inlet opening **553b**. When toner is supplied from the toner cartridge **570** to the developing device **590**, at least part of the inlet opening **553b** is positioned below at least part of the discharge opening **571b**.

(279) Then, toner stored in the toner storage portion **571a** is discharged through the discharge opening **571b**, and the toner discharged through the discharge opening **571b** is stored in the developing-side storage portion **553a** through the inlet opening **553b**. The toner stored in the developing-side storage portion **553a** is supplied to the developing roller **51** by the supply roller **52**. To be noted, the developing-side storage portion **553a** may include a toner conveyance member that conveys toner toward the supply roller **52**.

(280) The function of the toner cartridge **570** is substantially the same as the functions of the toner cartridge **70** of the first to fourth embodiments. In addition, the function of the developing device **590** is substantially the same as the function of one of the developing units **50y**, **50m**, **50c**, and **50k**

of the first to fourth embodiments.

(281) Meanwhile, the apparatus body **1A** includes a transfer roller **512**. The transfer roller **512** is an example of a transfer means or a transfer unit that transfers an image from the photosensitive drum **2** onto the sheet **S**. A conveyance roller pair **320** conveys the sheet **S** to a transfer portion that is a nip portion between the photosensitive drum **2** and the transfer roller **512**. The image on the photosensitive drum **2** is transferred onto the surface of the sheet **S** that has been conveyed thereto.

(282) The apparatus body **1A** includes a moving device configured to move the toner cartridge **570** from the attached position to the retracted position with respect to the developing device **590**, more specifically with respect to the developing frame **553** of the developing device **590**. As this moving device, a moving device described in the first to fourth embodiments and modification examples thereof can be used.

(283) In this case, among the moving devices of the first to fourth embodiments, part provided in the rotary body **90** may be provided in the developing device **590**. In addition, the replacement posture and the developing posture of the developing device **590** may be the same or different. For example, the developing device **590** may be movable to a contact position where the developing roller **51** is in contact with the photosensitive drum **2** and a separation position where the developing roller **51** is separated from the photosensitive drum **2**, and the developing device **590** may be at the separation position in a state in which the developing device **590** is in the replacement posture.

(284) For example, the developing device **590** may have the tray **80** and an element (moving device) for moving the tray **80**. As the element for moving the tray **80**, those described in the first to fourth embodiments and modification examples thereof can be used. In addition, the developing device **590** may include the rotary member **494a** and the driven roller **494b** that have been described in the modification example of the first embodiment.

(285) That is, the apparatus body **1A** of the present embodiment may include a moving device including a support member (tray **80**) that supports the toner cartridge **570**. In this case, the opening/closing member (door **14**) moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **570** from the attached position to the retracted position by the movement of the support member. As a result of this, the user does not have to open the door **14** manually, which makes it easier for the user to replace the toner cartridge.

(286) Also in the present embodiment, when the toner cartridge **570** is at the retracted position, at least part of the toner cartridge **570** is preferably outside the apparatus of the image forming apparatus **501** (outside the apparatus body **1A**). That is, when the toner cartridge **570** is at the retracted position, at least part of the toner cartridge **570** is positioned more outward than the exterior position with respect to the apparatus body **1A** when the toner cartridge **570** is at the retracted position. In other words, at least part of the toner cartridge **570** is positioned in a space outside the apparatus body **1A** in the case where the door **14** is at the closed position. Further, at least part of the toner cartridge **570** is positioned downstream of the exterior position in the retraction direction of the toner cartridge **570**.

(287) In addition, in the case where the side surface **16b** having the opening **16a** is the front surface of the apparatus body **1A**, it can be said at least part of the toner cartridge **570** projects to the front side more than the exterior surface on the front side of the apparatus body **1A** when the toner cartridge **570** is at the retracted position.

(288) Also in the present embodiment, when the toner cartridge **570** is at the retracted position, half the length or more of the toner cartridge **570** in the retraction direction is preferably outside the apparatus.

(289) As described above, although the toner cartridges **70** are detachably attached to the rotary body **90** in the first to fourth embodiments and modification examples thereof, the toner cartridge **570** is detachably attached to the developing device **590** in the present embodiment.

Other Embodiments

(290) In the embodiments described above, a configuration in which the tray **80** moves from the accommodation position to the detachment position by a driving device, an urging member, or the like of the apparatus body **1A** without the user operating the tray **80** has been described. Instead of this, a configuration in which the user grips the tray **80** and pulls out the tray **80** from the accommodation position to the detachment position in a state in which the rotary body **90** is at the replacement posture, that is, a configuration in which the support member is manually moved may be employed. Also in this case, since the user is only required to attach or detach the toner cartridge **70** to or from the tray **80** pulled out of the apparatus body, the operability is improved as compared with a case where the toner cartridge is directly inserted into or removed from the apparatus body.

(291) In addition, in the first to fourth embodiments described above, a configuration in which the rotary body **90** includes the four developing units **50y** to **50k** and a color image can be formed by using toners of four colors has been described. However, the number of developing units that the rotary body **90** includes may be 3 or less or 5 or more. In these cases, the number and arrangement of the trays and toner cartridges can be appropriately changed in accordance with the number of developing units. For example, in the first to fourth embodiments described above, a configuration in which the four toner cartridges **70y** to **70k** are attachable to and detachable from the rotary body **90** has been described as an example. However, a configuration in which the rotary body **90** includes only one developing unit **50k** and only one toner cartridge **70k** is attached to the rotary body **90** may be employed. In this case, the rotary body **90** can rotate in a clockwise direction in FIG. **1** about the rotational axis **90C** to alternately take the black replacement posture and the black developing posture.

(292) In addition, in the first to fourth embodiments described above, a configuration in which the rotary body **90** includes the four developing units **50y** to **50k** and a color image can be formed by using toners of four colors has been described. However, the rotary body **90** may include a plurality of developing units capable of executing image formation by using toner of the same color. For example, a configuration in which the rotary body **90** includes four black developing units **50k** and four toner cartridges **70k** are attached to the rotary body **90** may be employed.

(293) In addition, although a configuration in which the door **14** serving as an opening/closing member pivots about an axis substantially parallel to the rotational axis direction of the rotary body **90** (Y direction) has been described in the embodiments described above, the mode of movement of the door **14** is not limited to this. For example, a double door-like configuration in which the door **14** is split into two at the center thereof in the Y direction and the two parts of the door **14** respectively pivot about support shafts provided at respective end portions of the door **14** in the Y direction may be employed. In addition, for example, a configuration in which the door **14** is provided so as to be slidable in the Z direction with respect to the frame **16**, and the door **14** slides downward by being pushed by an inclined surface provided on the tray **80** in the case where the toner cartridge **70** moves in the retraction direction may be employed.

(294) According to the present disclosure, a new form of an image forming apparatus that advances conventional technology can be provided.

(295) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(296) This application claims the benefit of Japanese Patent Application Applications No. 2022-178517, filed on Nov. 8, 2022, and No. 2023-165975, filed on Sep. 27, 2023, which are hereby incorporated by reference herein in their entirety.

Claims

1. An image forming apparatus comprising: a rotary configured to rotate and including a developing roller and a storage frame including a storage portion configured to store toner to be supplied to the developing roller; a body frame with an opening, the body frame being configured to accommodate the rotary; a toner cartridge which is configured to store the toner, which is attachable to and detachable from the rotary through the opening, which is movable to an attached position and a retracted position where the toner cartridge is retracted from the attached position, with respect to the storage frame, and which is configured to supply the toner to the storage portion in a state in which the toner cartridge is at the attached position; a moving device configured to move the toner cartridge from the attached position to the retracted position; and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the moving device.
2. The image forming apparatus according to claim 1, wherein the moving device includes a support member which is configured to support the toner cartridge, to which the toner cartridge is detachably attached, and which is attached to the rotary, and wherein the support member is movable with respect to the storage frame such that the toner cartridge moves from the attached position to the retracted position.
3. The image forming apparatus according to claim 2, wherein in a case where the moving device moves the toner cartridge from the attached position to the retracted position, the support member is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position.
4. The image forming apparatus according to claim 2, further comprising: an urging member configured to urge the opening/closing member toward the closed position, wherein the opening/closing member is configured to be moved from the open position to the closed position by an urging force of the urging member in an interconnected manner with movement of the toner cartridge from the retracted position to the attached position by the moving device.
5. The image forming apparatus according to claim 4, wherein the opening/closing member is pivotable to the open position and the closed position about a rotational axis extending in a direction intersecting with a direction in which the support member moves.
6. The image forming apparatus according to claim 1, wherein the body frame includes an exterior surface constituting an exterior of the image forming apparatus, wherein the opening is provided in the exterior surface, and wherein in a case where the opening/closing member is at the closed position, the opening/closing member constitutes part of the exterior surface.
7. The image forming apparatus according to claim 6, wherein the toner cartridge is configured such that, in a state in which the toner cartridge is at the retracted position, at least part of the toner cartridge projects outside the body frame with respect to a position of the exterior surface of a case where the opening/closing member is at the closed position.
8. The image forming apparatus according to claim 1, wherein the body frame has a first surface constituting part of an exterior surface of the image forming apparatus and a second surface provided at a more inward position of the image forming apparatus than the first surface, and wherein the opening is provided in the second surface.
9. The image forming apparatus according to claim 8, further comprising: a sheet storage portion which is configured to store a sheet and which is movable, with respect to the body frame, between a stored position where the sheet storage portion is stored in the body frame and a drawn-out position where the sheet storage portion is drawn out from the stored position in a draw-out

direction such that at least part of the sheet storage portion projects outside the body frame, wherein in a state in which the sheet storage portion is at the stored position, a surface of the sheet storage portion on a downstream side in the draw-out direction, together with the first surface, constitutes an exterior surface of the image forming apparatus on the downstream side in the draw-out direction.

10. The image forming apparatus according to claim 1, further comprising: a second opening/closing member movable to a position where the second opening/closing member covers the opening/closing member at the closed position and a position where the second opening/closing member exposes the opening/closing member so as to allow movement of the opening/closing member from the closed position to the open position.

11. The image forming apparatus according to claim 1, wherein the opening/closing member is configured such that in a state in which the toner cartridge is at the retracted position and the opening/closing member is at the open position, at least part of a weight of the toner cartridge is supported by the opening/closing member.

12. The image forming apparatus according to claim 1, wherein a movement direction of the toner cartridge from the attached position toward the retracted position is a direction intersecting with a rotational axis direction of the rotary.

13. The image forming apparatus according to claim 1, further comprising: a photosensitive drum, wherein the rotary is rotatable to a developing posture in which the developing roller faces the photosensitive drum and a replacement posture where detachment of the toner cartridge from the rotary is allowed.

14. The image forming apparatus according to claim 1, wherein in a case where the moving device moves the toner cartridge from the attached position to the retracted position, the toner cartridge is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position.

15. The image forming apparatus according to claim 14, wherein the moving device includes a support member which is configured to support the toner cartridge, to which the toner cartridge is detachably attached, and which is attached to the rotary, and wherein the support member is movable with respect to the storage frame such that the toner cartridge moves from the attached position to the retracted position.

16. The image forming apparatus according to claim 15, further comprising: an urging member configured to urge the opening/closing member toward the closed position, wherein the opening/closing member is configured to be moved from the open position to the closed position by an urging force of the urging member in an interconnected manner with movement of the toner cartridge from the retracted position to the attached position by the moving device.

17. The image forming apparatus according to claim 16, wherein the opening/closing member is pivotable to the open position and the closed position about a rotational axis extending in a direction intersecting with a direction in which the support member moves.

18. The image forming apparatus according to claim 14, wherein the body frame has a first surface constituting part of an exterior surface of the image forming apparatus and a second surface provided at a more inward position of the image forming apparatus than the first surface, and wherein the opening is provided in the second surface.

19. The image forming apparatus according to claim 18, further comprising: a sheet storage portion which is configured to store a sheet and which is movable, with respect to the body frame, between a stored position where the sheet storage portion is stored in the body frame and a drawn-out position where the sheet storage portion is drawn out from the stored position in a draw-out direction such that at least part of the sheet storage portion projects outside the body frame, wherein in a state in which the sheet storage portion is at the stored position, a surface of the sheet storage portion on a downstream side in the draw-out direction, together with the first surface, constitutes an exterior surface of the image forming apparatus on the downstream side in the draw-

out direction.

20. An image forming apparatus comprising: a rotary including a first developing roller, a second developing roller, a first storage frame, and a second storage frame, the first storage frame including a first storage portion configured to store first toner to be supplied to the first developing roller, the second storage frame including a second storage portion configured to store second toner to be supplied to the second developing roller; a body frame configured to accommodate the rotary and having an opening; a first toner cartridge which is configured to store the first toner, which is attachable to and detachable from the rotary through the opening, which is movable to a first attached position and a first retracted position where the first toner cartridge is retracted from the first attached position, with respect to the first storage frame, and which is configured to supply the first toner to the first storage portion in a state in which the first toner cartridge is at the first attached position; a second toner cartridge which is configured to store the second toner, which is attachable to and detachable from the rotary through the opening, which is movable to a second attached position and a second retracted position where the second toner cartridge is retracted from the second attached position, with respect to the second storage frame, and which is configured to supply the second toner to the second storage portion in a state in which the second toner cartridge is at the second attached position; a first moving device configured to move the first toner cartridge from the first attached position to the first retracted position; and a second moving device configured to move the second toner cartridge from the second attached position to the second retracted position; and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the first toner cartridge and the second toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the first toner cartridge from the first attached position to the first retracted position by the first moving device, and wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the second toner cartridge from the second attached position to the second retracted position by the second moving device.

21. The image forming apparatus according to claim 20, wherein the first moving device includes a first support member which is configured to support the first toner cartridge, to which the first toner cartridge is detachably attached, and which is attached to the rotary, wherein the second moving device includes a second support member which is configured to support the second toner cartridge, to which the second toner cartridge is detachably attached, and which is attached to the rotary, wherein the first support member is movable with respect to the first storage frame such that the first toner cartridge moves from the first attached position to the first retracted position, and wherein the second support member is movable with respect to the second storage frame such that the second toner cartridge moves from the second attached position to the second retracted position.

22. The image forming apparatus according to claim 21, wherein in a case where the first moving device moves the first toner cartridge from the first attached position to the first retracted position, the first support member is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position, wherein in a case where the second moving device moves the second toner cartridge from the second attached position to the second retracted position, the second support member is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position.

23. The image forming apparatus according to claim 20, wherein the opening/closing member is configured such that in a state in which the first toner cartridge is at the first retracted position and the opening/closing member is at the open position, at least part of a weight of the first toner

cartridge is supported by the opening/closing member, and wherein the opening/closing member is configured such that in a state in which the second toner cartridge is at the second retracted position and the opening/closing member is at the open position, at least part of a weight of the second toner cartridge is supported by the opening/closing member.

24. The image forming apparatus according to claim 20, wherein in a case where a length by which the first toner cartridge at the first retracted position projects from the opening is a first length and a length by which the second toner cartridge at the second retracted position projects from the opening is a second length, the first length is larger than the second length.

25. The image forming apparatus according to claim 20, wherein a movement amount of the first toner cartridge from the first attached position to the first retracted position is larger than a movement amount of the second toner cartridge from the second attached position to the second retracted position.

26. The image forming apparatus according to claim 20, wherein the first toner cartridge is configured to store a first amount of the first toner, wherein the second toner cartridge is configured to store a second amount of the second toner, and wherein the first amount is larger than the second amount.

27. The image forming apparatus according to claim 20, wherein a movement direction of the first toner cartridge in which the first toner cartridge moves from the first attached position to the first retracted position intersects with a rotational axis direction of the rotary, and wherein a movement direction of the second toner cartridge in which the second toner cartridge moves from the second attached position to the second retracted position intersects with the rotational axis direction of the rotary.

28. The image forming apparatus according to claim 20, further comprising: a photosensitive drum, wherein the rotary is rotatable to a first developing posture in which the first developing roller faces the photosensitive drum, a first replacement posture in which detachment of the first toner cartridge from the rotary is allowed, a second developing posture in which the second developing roller faces the photosensitive drum, and a second replacement posture in which detachment of the second toner cartridge from the rotary is allowed.
